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Paleomagnetic constraints on the Archean geomagnetic field intensity obtained from komatiites of the Barberton and Belingwe greenstone belts, South Africa and Zimbabwe

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Abstract

The intensity of the geomagnetic field during Archean is an important source of information about the evolution of the Earth's core. Hale [Earth Planet. Sci. Lett. 86 (1987b) 354] reported a very low equatorial paleointensity of 5 μT at ca. 3.5 Ga obtained from komatiites of the Barberton greenstone belt, South Africa, and regarded their remanences as thermal overprints due to a metamorphic event. However, this result has a major drawback since remanences of the Barberton komatiites are carried by magnetites formed during serpentinization that typically results in acquisition of chemical remanent magnetization (CRM). We report the results of detailed paleomagnetic, rock magnetic, and paleointensity investigations using komatiite samples from the Barberton (ca. 3.5 Ga) and Belingwe (ca. 2.7 Ga) greenstone belts in South Africa and Zimbabwe, respectively. Paleodirectional data for Barberton are consistent with results of previous work, and the characteristic remanent magnetization (ChRM) of the Belingwe samples reveals a positive fold test, suggesting that samples from both areas record the Archean geomagnetic field. The main carriers of ChRM are Ti-free magnetites presumed to be akin to single domain grains, and their rock magnetic properties are very stable when subjected to laboratory heating up to 600–700 °C. Our microscopic observations provide support that these are secondary minerals formed during serpentinization. Thellier–Thellier paleointensity experiments for the Barberton and Belingwe samples yielded very low mean virtual dipole moment (VDM) estimates of $(1.8 \pm 1.3) \times 10^{22}$ A m^2 and $(1.1 \pm 0.9) \times 10^{22}$ A m^2, respectively. These correspond to about 24 and 15% of the present day value. Considering the low metamorphic grade for the sampling areas, however, it seems to be somewhat difficult to regard the entire ranges of komatiite ChRM as thermal overprints acquired by reheating during metamorphic events. We hence propose as the most likely scenario the possibility that most fractions of ChRM are survivors of primary grain growth CRM. Taking the theoretical and experimental estimates of the ratio of CRM to thermoremanent magnetization (TRM) into account, this suggests that these low mean VDM values can provide constraints on the lower limits of the geomagnetic field intensities in both greenstone belts at ca. 3.5 and 2.7 Ga.
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Keywords: Paleointensity; Archean; Komatiite; Barberton greenstone belt; Belingwe greenstone belt

1. Introduction

The long-term variation in magnitude of the Earth's magnetic field provides an important source of information about the dynamics of the Earth's core. In particular, the paleointensity variation during Archean

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and Proterozoic times is considered to reflect mode changes in powering the geodynamo. Some previous studies on the Earth's thermal history suggest that the onset of inner core nucleation occurred during these times (Stevenson et al., 1983; Buffett et al., 1992; Labrosse et al., 2001). Using a simple dynamo model, Stevenson et al. (1983) revealed that the release of gravitational energy and latent heat that accompanied the inner core formation drastically increased the field intensity. Hale (1987a) recognized a sharp increase in magnitude of the geomagnetic field between 2.7 and 2.1 Ga in his compilation of paleointensity data and suggested a link between the onset of inner core nucleation and the Archean–Proterozoic boundary. Breuer and Spohn (1995) presented simulations of mantle convection at the end of the late Archean and showed that a breakdown of two-layer convection at the Archean–Proterozoic boundary also brought about a significant increase in field intensity due to enhancement of core cooling. Although some recently published estimates of paleointensities during the late Archean seem to be consistent with the picture of steadily increasing field intensity around the Archean–Proterozoic boundary (Morimoto et al., 1997; Yoshihara and Hamano, 2000; Selkin et al., 2000), the volume of reliable data is insufficient to examine the analyses using these models. Especially for the early to middle Archean, we have only scant information about the geomagnetic field.

Paleointensity data giving the oldest age published to date were obtained from komatiites of the Barberton greenstone belt, South Africa, with an age of ca. 3.5 Ga (Hale and Dunlop, 1984; Hale, 1987b) and are the only results for the early to middle Archean. Using the best preserved komatiites from the Komati Formation of the Barberton greenstone belt, Hale and Dunlop (1984) obtained characteristic remanences that yielded well clustered paleomagnetic directions and concluded that the remanences were overprinted thermoremanent magnetization (TRM) acquired during greenschist metamorphism at ca. 3.5 Ga. Hale (1987b) performed paleointensity experiments on the same samples by the Thellier–Thellier method (Thellier and Thellier, 1959) and the van Zijl method (van Zijl et al., 1962a,b), and obtained a mean equatorial paleointensity of $5 \pm 2\,\mu$T, which corresponds to 17% of the present day value. There are, however, some shortcomings to Hale's (1987b) findings. The major drawback is that the remanences of the komatiite samples used are carried by magnetite formed during serpentinization that typically results in acquisition of chemical remanent magnetization (CRM) as pointed out by Prévot and Perrin (1992). The procedures and analyses in Hale's experiments appear not to satisfy the common criteria currently used for Thellier–Thellier type paleointensity experiments in that no consistency check was applied, e.g., the partial TRM (pTRM) test (Coe, 1967). Moreover, it seems necessary to assess the magnetic properties of the komatiite samples in greater detail to ensure the significance of the paleointensity results.

Here, we report results of detailed paleomagnetic, rock magnetic, and Thellier–Thellier paleointensity investigations with Archean komatiites. Used in the present study were komatiites occurring in the Barberton greenstone belt, South Africa, and in the Belingwe greenstone belt, Zimbabwe. The previous paleointensity report by Hale and Dunlop (1984) and Hale (1987b) were reexamined using samples from the same locality in Barberton. In the Belingwe greenstone belt, komatiites are as well preserved as those of Barberton, having a much younger age of ca. 2.7 Ga, but paleomagnetic and paleointensity studies on this greenstone belt have not been performed. Paleomagnetic and rock magnetic properties are compared between these two greenstone belts, and the origin of the komatiite remanences and the validity of these paleointensity results are discussed.

2. Geological setting and sampling

2.1. Komatiites of the Barberton greenstone belt

The Barberton Mountain Land located in the Kaapvaal craton near its eastern margin (Fig. 1) is one of the oldest and best preserved of the Archean granite–greenstone terranes. The Barberton greenstone belt, which is contained within the Barberton Mountain Land, is composed of metavolcanic rocks and metasediments surrounded by a variety of granitoid plutons and is strongly folded synclinally with a dominant ENE-trend (Viljoen and Viljoen, 1969a; Kröner et al., 1996) (Fig. 2). Recent zircon geochronology has established an almost uninterrupted record of mafic–ultramafic and felsic mag-

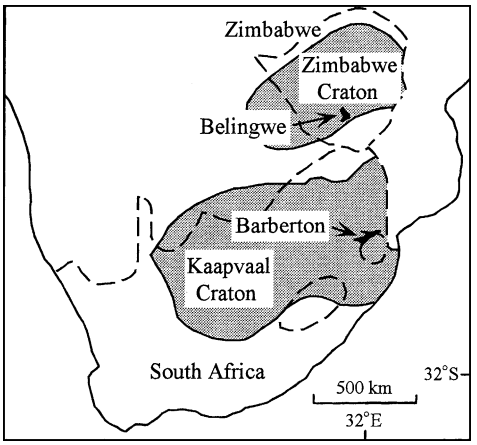


Fig. 1. Location of the Barberton and Belingwe greenstone belts.

matism over the period from ca. 3550 to 3230 Ma (Armstrong et al., 1990; Kröner et al., 1991, 1996; López-Martínez et al., 1992; Kamo and Davis, 1994; de Ronde and Kamo, 2000).

The Onverwacht Group is the lowest, mainly volcanic succession of over 15 km thick, and overlain by the sedimentary-volcanic Fig Tree Group and the sedimentary Moodies Group (Viljoen and Viljoen, 1969a) (Fig. 2). Some early Archean crustal evolution scenarios have used this greenstone belt as a model (Eriksson, 1980; Kröner, 1985; Condie, 1989). Recently, this greenstone belt has been interpreted as an accretionary complex of an Archean oceanic crust (de Wit et al., 1992). Thus, its volcanic rocks are mainly composed of mid-oceanic ridge basalts and oceanic flood basalts of hot-spot origin. Particularly, komatiites and their differentiated volcanics in the greenstones probably correspond to hot-spot volcanic rocks (Komiya and Maruyama, personal communication, 2000).

The target of the present investigation is komatiites exposed in the best preserved type locality area of the Komati Formation in the Onverwacht Group (Fig. 3). The Komati Formation predominantly consists of high-MgO peridotitic komatiites (olivine-dominated,

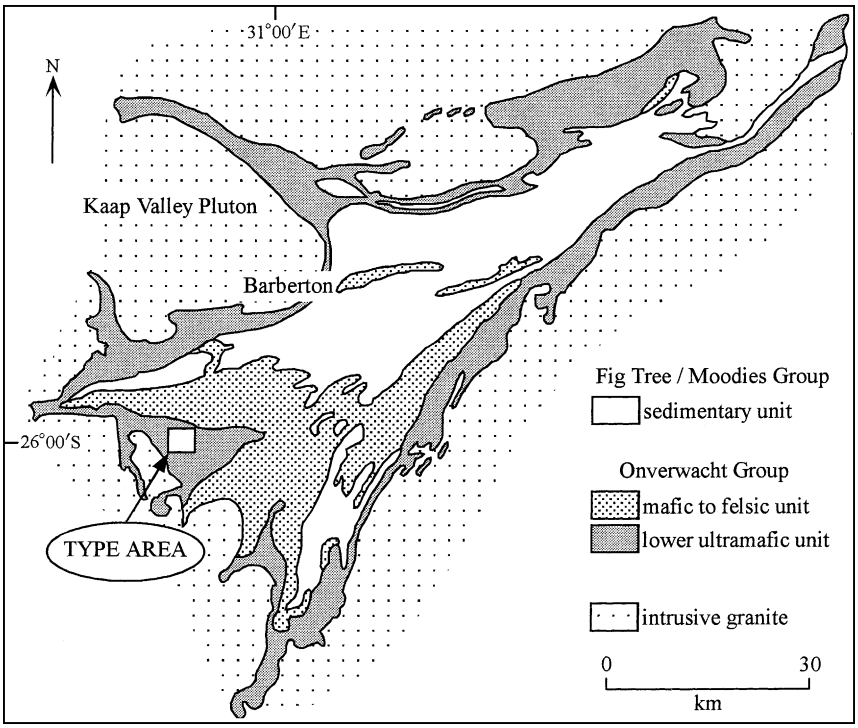


Fig. 2. Geological map of the Barberton greenstone belt (simplified from Viljoen and Viljoen, 1969b). The type area of the Komati Formation is indicated.

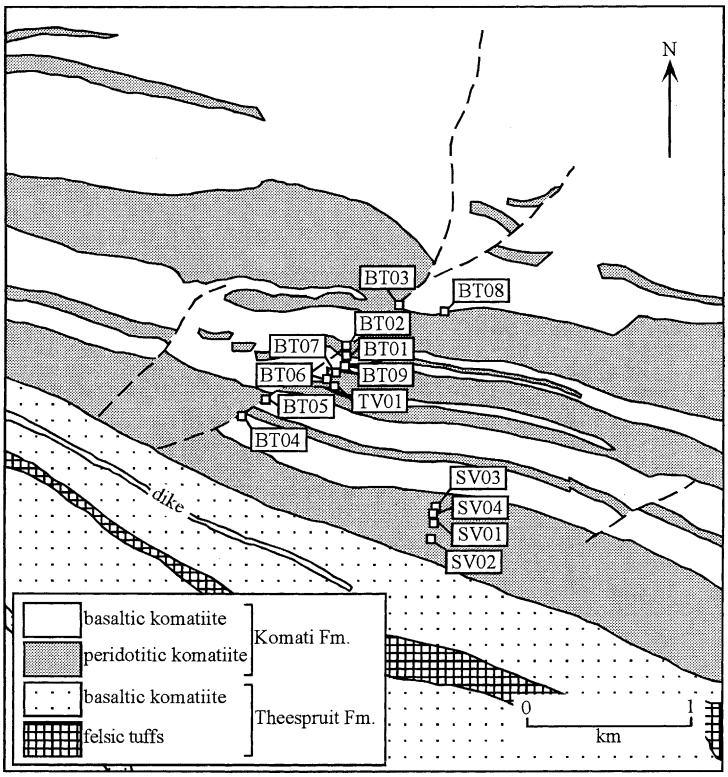


Fig. 3. Geological map of the type area of the Komati Formation (simplified from Viljoen and Viljoen, 1969b). Localities of the sampling sites are indicated.

massive and spinifex-textured) interlayered with low-MgO basaltic komatiites (pyroxene-dominated, massive and pillowed) with minor metatholeiites and small intrusive bodies, forming a steeply dipping, NNE-younging flow sequence (Viljoen and Viljoen, 1969b; Cloete, 1994). The peridotitic komatiites are remarkably fresh-looking, fine-grained dense rocks, with metamorphism usually in the greenschist facies (Cloete, 1990; Komiya, 1999). However, microscopic observations reveal extensive alteration and serpentinization with only rarely preserved kernels of olivine in a groundmass composed of antigolite, tremolite, chlorite and magnetite (Viljoen and Viljoen, 1969b).

According to Kamo and Davis (1994), the age of the lower Komati Formation is bracketed between 3458 ± 2 Ma, the U–Pb age determined for a felsic dike that intrudes into the formation (Kamo and Davis, 1994), and 3472 ± 5 Ma, the U–Pb age for a volcaniclastic interflow at about the same stratigraphic level (Armstrong et al., 1990). The komatiites of the Komati Formation are generally of low metamorphic grade, but have evidently been affected by pervasive seafloor metamorphism (de Wit et al., 1987). López-Martínez et al. (1992) reported an Ar–Ar metamorphic age of 3486 ± 8 Ma for the komatiites in this area. This is slightly older than the U–Pb age constraint for the komatiites, and Kamo and Davis (1994) argued that alteration seems to have occurred somewhere in the time span between 3450 and 3490 Ma, based on the broad consistency of the Ar–Ar ages.

Peridotitic komatiites and basaltic komatiites of the lower Komati Formation were collected at 14 sites in the type locality area (Fig. 3). Table 1 shows details of the sampling sites. Sites BT01–09 and SV01–04 are distributed along small streams, and only site TV01 is located on a hillside. The paleointensities reported by Hale and Dunlop (1984) and Hale (1987b) were also obtained from komatiites of this type locality area. In particular, the outcrops of

Table 1
Sampling sites, NRM intensities and Q_n ratios of komatiites from the Barberton greenstone belt

Stratigraphy	Site	Latitude (S)	Longitude (E)	Lithology	NRM (A/m)	Q_n ratio
Komati Formation	BT02	25°59.25′	30°51.61′	Peridotitic	8–24	8–25
	BT06	25°59.31′	30°51.51′	Peridotitic	7–13	4–6
	BT07	25°59.30′	30°51.51′	Peridotitic	1–19	1–12
	BT09	25°59.27′	30°51.58′	Peridotitic	4–12	2–5
	SV01	25°59.79′	30°52.16′	Peridotitic	5–9	2–4
	SV02	25°59.82′	30°52.15′	Peridotitic	2–7	1–4
	SV04	25°59.73′	30°52.16′	Peridotitic	4–11	2–6
	TV01	25°59.33′	30°51.53′	Peridotitic	1–9	2–6
	BT01	25°59.26′	30°51.59′	Basaltic	–	–
	BT03	25°59.12′	30°51.78′	Basaltic	2×10^{-4} to 6×10^{-4}	<0.1
	BT04	25°59.42′	30°51.20′	Basaltic	1×10^{-4} to 4×10^{-4}	<0.1
	BT05	25°59.34′	30°51.38′	Basaltic	1×10^{-4} to 2×10^{-4}	<0.1
	BT08	25°59.08′	30°51.92′	Basaltic	–	–
	SV03	25°59.70′	30°52.16′	Basaltic	2×10^{-5} to 3×10^{-4}	<0.1

sites BT01–09 are identical or very close to those at which they sampled. At each site, we collected 5–12 drilled or block samples oriented by a magnetic compass, carefully avoiding sampling from such strongly magnetized blocks as cause incorrect magnetic orientation.

2.2. Komatiites of the Belingwe greenstone belt

The Belingwe greenstone belt is located in the southeast portion of the Zimbabwe craton that forms the central part of Zimbabwe (Fig. 1), and contains lower and upper sequences of greenstones termed the Mtshingwe Group and the Ngezi Group, respectively, being surrounded by a granite-gneiss terrane ranging from 3.6 to 2.6 Ga in age (Martin et al., 1993) (Fig. 4). Both the Mtshingwe and Ngezi Groups have relatively simple synclinal depositions, which contain a wide range of sedimentary and volcanic rock types (Bickle et al., 1993). The age of the Ngezi Group has been precisely dated. Basaltic komatiites occurring in the Reliance Formation of the Ngezi Group yield a Pb–Pb age of 2692 ± 9 Ma (Chauvel et al., 1993), which is consistent with earlier regional Sm–Nd and Rb–Sr isochron ages of 2640 ± 140 Ma (Hamilton et al., 1977) and 2700 ± 70 Ma (Jahn and Condie, 1976), respectively. While the age of the Mtshingwe Group has not been well-defined, it is probably similar to the 2.9–2.8 Ga of the Chingezi Gneiss (Martin et al., 1993). The Ngezi Group forms the central north–northwest trending syncline of the belt, but because this first deformation produced little internal strain, many primary volcanic and sedimentary textures are almost perfectly preserved (Martin et al., 1993). Since the Chilimanzi Suite with an age of 2570 ± 25 Ma (Hickman, 1978) clearly crosscuts parts of the greenstone belt and has been little deformed, this folding must predate this granite intrusion and therefore occurred between 2692 and 2570 Ma (Bickle et al., 1993).

Two scenarios have been proposed to illustrate the development of the greenstone belt. One is the continental extensional setting (Blenkinsop et al., 1993; Bickle et al., 1994; Hunter et al., 1998), and the other is the oceanic plateau setting (Kusky and Kidd, 1992; Kusky and Winsky, 1995). The existence of basal conglomerates supports the development of the greenstone belt in a continental rift setting. Thus, the komatiites and their differentiated volcanics in the greenstones probably correspond to continental hot-spot volcanic rocks (Komiya, personal communication, 2000).

The target of the present study is the best preserved komatiites with a Pb–Pb age of 2692 ± 9 Ma (Chauvel et al., 1993) occurring in the Reliance Formation of the Ngezi Group. The Reliance Formation consists predominantly of komatiites and basaltic komatiites, in which clinopyroxene and olivine spinifex textures are commonly developed throughout the for-

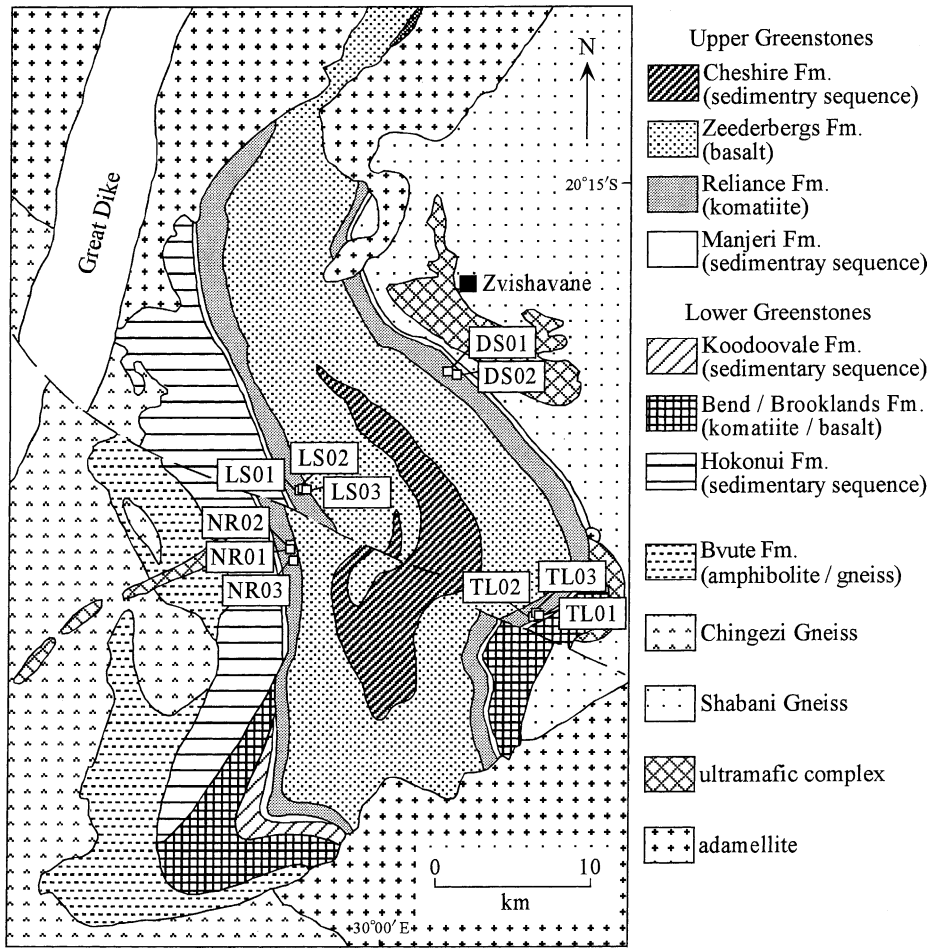


Fig. 4. Geological map of the Belingwe greenstone belt (simplified from Martin et al., 1993). Localities of the sampling sites are indicated.

mation (Martin et al., 1993). Metamorphism is usually in the low greenschist or sub-greenschist facies, but fresh olivine is less common and has been substantially or completely replaced by serpentine-magnetite or chlorite (Bickle et al., 1993).

Peridotitic komatiites and basaltic komatiites of the Reliance Formation were collected at 11 sites (Fig. 4). Details of the sampling sites are listed in Table 2. Sites TL01–03 are at the type section described by Bickle et al. (1975), and the localities of sites DS01–02 are very close to the SASKMAR borehole geologically investigated by Renner et al. (1994). At each site, we collected 8–20 samples in the same way as the Barberton case.

3. Paleomagnetic directions

To assess the stability of the natural remanent magnetization (NRM) possessed by the komatiite samples, NRM and Koenigsberger ratio (Q_n ratio) measurements and thermal demagnetization experiments were performed. Characteristic remanent magnetization (ChRM) components were isolated, and paleomagnetic mean directions were calculated for both formations. An AGICO spinner magnetometer, a Schonstead spinner magnetometer and a 2G SQUID magnetometer were used for the remanence measurements, and thermal demagnetization was conducted with a Natsuhara thermal demagnetizer. Initial mag-

Table 2
Sampling sites, NRM intensities and Q_n ratios of komatiites from the Belingwe greenstone belt

Stratigraphy	Site	Latitude (S)	Longitude (E)	Lithology	NRM (A/m)	Q_n ratio	Note
Reliance Formation	DS01	20°21.39′	30°02.39′	Peridotitic	1–5	1–6	Borehole locality[a]
	DS02	25°21.50′	30°02.70′	Peridotitic	2–6	2–4	Borehole locality[a]
	TL03	25°29.89′	30°05.58′	Peridotitic	1–4	1–2	Type locality[b]
	LS01	25°25.31′	29°57.01′	Peridotitic	1–100	3–360	
	LS02	25°25.35′	29°57.03′	Peridotitic	3–28	3–33	
	NR01	25°27.54′	29°56.80′	Peridotitic	1–2	3–9	
	NR02	25°27.42′	29°56.77′	Peridotitic	1–2	2–5	
	NR03	25°28.04′	29°56.93′	Peridotitic	1–2	2–4	
	TL01	25°29.95′	30°05.52′	Basaltic	–	–	Type locality[b]
	TL02	25°30.16′	30°05.59′	Basaltic	1×10^{-2} to 2×10^{-2}	1–2	Type locality[b]
	LS03	25°25.28′	29°57.19′	Basaltic	1×10^{-1} to 9×10^{-1}	5–49	

[a] Locality close to the borehole investigated by Renner et al. (1994).
[b] Type section described by Bickle et al. (1975).

netic susceptibility was measured with a Bartington susceptibility meter.

3.1. Results from komatiites of the Barberton greenstone belt

The peridotitic komatiite samples from the Komati Formation were strongly magnetized, having NRM intensities typically ranging from 1 to 20 A/m. On the other hand, the NRM values of the basaltic komatiite samples were very weak, representing intensity values in the order of 10^{-5} to 10^{-4} A/m. Since reliable remanence values were not obtained with the SQUID magnetometer for samples from sites BT01 and BT08, those samples were not used for the following investigations. Site-mean Q_n ratios for the peridotitic komatiites ranged from 3 to 13, suggesting their stable nature against presently induced magnetizations. Basaltic komatiites had much smaller Q_n values less than 0.1. In Fig. 5a, Q_n ratios were plotted versus NRM intensities. These two types of komatiite showed significant differences both in order of values and in trend, reflect-

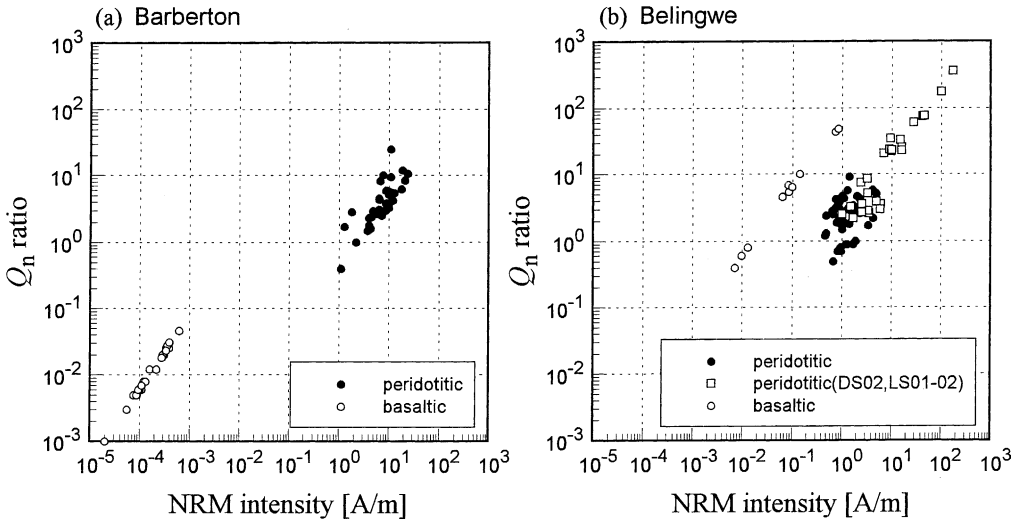


Fig. 5. Plots of Q_n ratios vs. NRM intensities for komatiite samples from the Barberton (a) and Belingwe (b) greenstone belts.

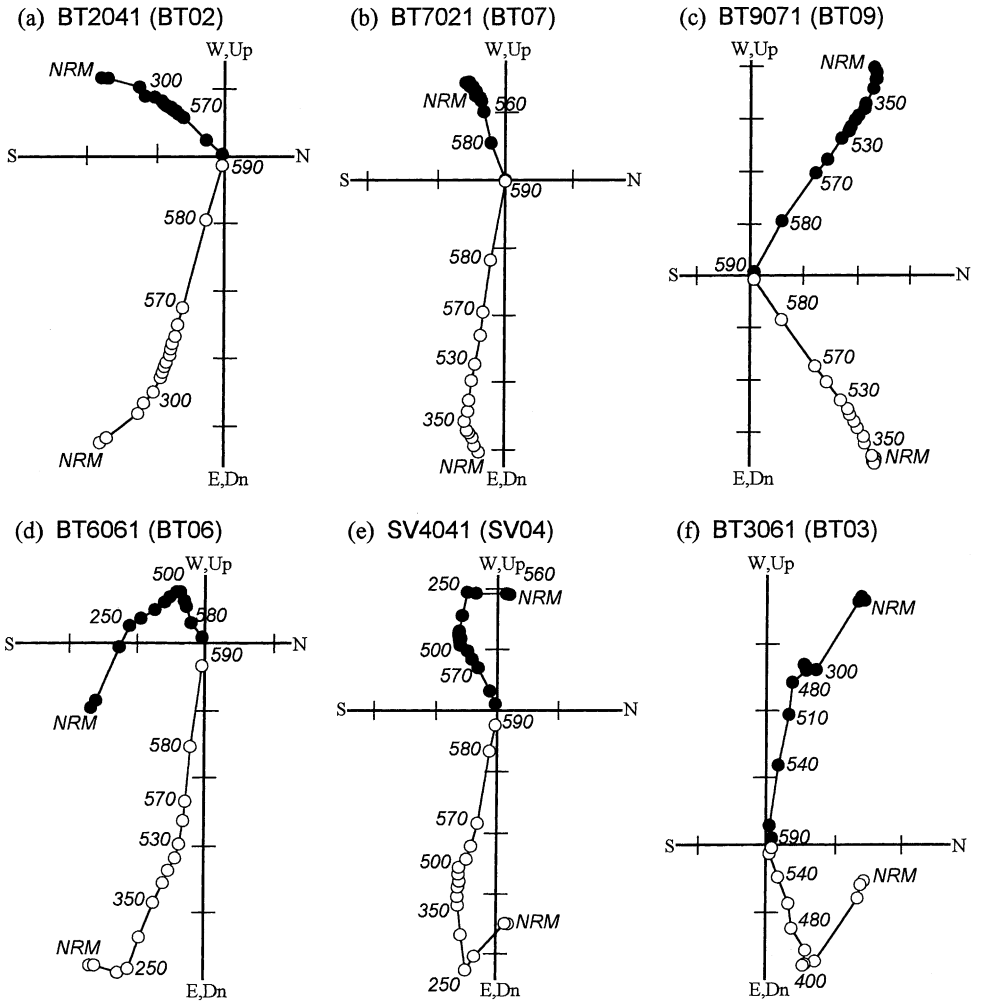


Fig. 6. Results of thermal demagnetization for peridotitic komatiite (a–e) and basaltic komatiite (f) samples from the Barberton greenstone belt. Open (solid) symbols represent vector endpoints projected onto the vertical (horizontal) plane. Numbers beside the points represent temperatures.

ing the petrological difference between them. Details of NRM intensities and Q_n ratios are summarized in Table 1.

Four to eight samples for each site were thermally demagnetized at temperatures up to 590 °C. Typical demagnetization curves for the peridotitic samples are shown in Fig. 6a–e. The NRM of most samples mainly consisted of two components (Fig. 6a–c). Low-temperature components that seem to be viscous remanences were removed by heating to 300 °C, and high-temperature components could be isolated between 300 and 590 °C. As can be seen in Fig. 6d and e, more complicated demagnetization curves were obtained for some samples, and middle-temperature components were observed at temperatures ranging between 300 and 500 °C. The most significant feature of the results from the peridotitic samples is that their blocking temperature distributions were extremely concentrated in the high temperature range, particularly the 570–590 °C range. For many samples, more than half of the initial NRM intensity remained after thermal demagnetization up to 570 °C, and then dras-

tically decayed by heating to 590 °C. This suggests that their NRM is mainly carried by fine-grained magnetites. As for the basaltic komatiites, stable components were isolated only from nine samples from sites BT03 and SV03. A successful example of thermal demagnetization is shown in Fig. 6f. Because of their weak magnetization, even in the successful cases, their demagnetization curves are not necessarily of high quality, disturbed by the background noise of the magnetometer to some extent. Nevertheless, at least two components can be recognized in these samples. One is the low-temperature component of viscous remanence removed by heating up to 400 °C and the other is the high-temperature component isolated in the range from 400–500 °C to 580 °C. Unlike observations on the peridotitic samples, extreme concentrations of the blocking temperature distribution to the high temperature range were not observed.

For the peridotitic komatiites, high-temperature components were isolated from 46 samples from seven sites as ChRM and their site-mean directions were calculated. Table 3 shows details of the directional data. Most showed steep inclinations and western-seeking declinations, and were well clustered within each site. However, the directions of samples from site TV01 had northeastern-seeking declinations and shallower inclinations. While the other sites are distributed along the small streams, only site TV01 is located on the hillside. The strata at site TV01 might have a different bedding from the other sites due to the collapse of the outcrop. Therefore, the mean direction of site TV01 was not used for calculating the formation-mean direction. Fig. 7a shows the site-mean directions of the high-temperature components with their 95% confidence limits (except for site TV01). The in situ formation-mean direction is $D = -87.7°$, $I = 72.6°$, with $\alpha_{95} = 12.0°$, which closely agrees with the result reported by Hale and Dunlop (1984) as shown in Fig. 7b. Hale and Dunlop (1984) did not make a structural correction for their mean in situ direction, since they thought that the characteristic remanences are post-kinematic remanences, i.e., overprinted TRMs. We calculated tilt-corrected mean directions. As will be discussed in Section 7, there is the possibility that the high-temperature components are primary remanences acquired soon after the extrusions, i.e., pre-tilting remanences. For the correction, values of strike and dip measured at site BT09 were

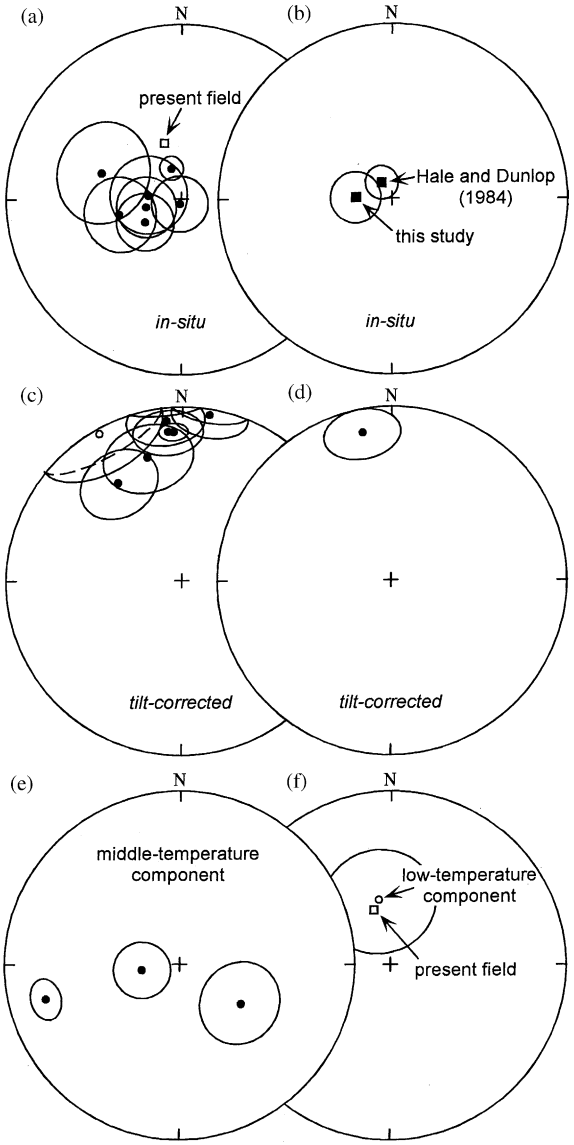


Fig. 7. Mean paleomagnetic directions of komatiite samples from the Barberton greenstone belt with 95% confidence limits. Solid (open) symbols are plotted on the lower (upper) hemisphere by an equal area projection. (a) In situ site-mean directions. (b) In situ formation-mean direction compared to the result by Hale and Dunlop (1984). (c) Tilt-corrected site-mean directions. (d) Tilt-corrected formation-mean direction. (e) Site-mean directions of the middle-temperature components. (f) Mean direction of the low-temperature components.

Table 3
Paleomagnetic directions of komatiites from the Barberton greenstone belt

Site	Bedding	n	In situ		Tilt-corrected		α_{95} (°)	k
	Strike/dip (°)		Dec (°)	Inc (°)	Dec (°)	Inc (°)		
High-temperature component								
BT02	280/88	5	237.8	69.8	354.6	15.3	13.5	33.1
BT06	280/88	6	202.4	87.7	9.5	4.3	13.5	25.7
BT07	280/88	5	257.2	73.1	354.3	8.4	14.2	29.8
BT09	280/88	4	288.6	50.4	330.8	−3.9	22.8	17.3
SV01	272/60	8	277.2	74.5	344.6	27.4	18.5	10.0
SV02	272/60	7	341.9	74.8	356.6	15.6	5.7	112.1
SV04	272/60	6	255.9	60.3	327.2	33.6	17.3	16.0
TV01	280/88	5	48.5	47.8	38.9	−30.0	24.0	11.1
Mean(in situ)[a]		7	272.3	72.6			12.0	26.3
Mean(tilt-corrected)[a]		7			348.8	14.8	14.8	17.6
Middle-temperature component								
BT06	280/88	6	122.7	55.6			19.3	13.0
SV02	272/60	7	255.3	22.2			8.4	52.8
SV04	272/60	3	261.7	72.3			13.4	85.7
Low-temperature component								
Mean(in situ)[b]		48	349.9	−58.8			25.5	1.6

n, number of samples used for paleodirectional determinations; Dec, mean declination; Inc, mean inclination; α_{95}, 95% confidence limit; k, Fisherian precision parameter.
[a] Mean direction calculated with site-means except for the result from TV01.
[b] Mean direction calculated with directions from all samples.

used as a bedding for sites BT02, BT06, BT07 and BT09. Tilting of sites SV01, SV02 and SV03 was corrected with the values for strike and dip measured at site SV01. The tilt-corrected formation-mean direction was $D = -11.2°$, $I = 14.8°$, with $\alpha_{95} = 14.8°$ (Fig. 7c and d). By application of the tilt correction, the radius of the 95% confidence limit of the mean direction, α_{95}, slightly increased from 12.0 to 14.8°, and the Fisherian precision parameter, k, decreased from 26.3 to 17.6. However, any fold test seems inconclusive because only two representative bedding data were employed for all sampling sites where the bedding attitudes are very similar.

The middle-temperature components were isolated from samples from sites BT06, SV02 and SV04. As shown in Fig. 7e, their directions were well clustered within each site. However, the site-mean directions of these three sites did not yield a characteristic direction between sites. Although interpreting this divergence of the site-mean directions presents some difficulty, their temperature ranges seem consistent with the temperature condition of greenschist metamorphism. The low-temperature components were isolated from 48 samples, and the mean direction for all samples is $D = -10.1°$, $I = -58.8°$, with $\alpha_{95} = 25.5°$ (Fig. 7f). This direction is very similar to the present geomagnetic field direction in the Barberton region ($D = -17.6°$, $I = -62.8°$) calculated from IGRF, 2000 (IAGA Division V Working Group 8, 2000), strongly suggesting that the low-temperature components are viscous remanent magnetizations acquired during exposure to a recent field.

As for the basaltic komatiites from sites BT03 and SV03, the directions of their high-temperature components were well-clustered within each site, and both site-mean directions revealed much shallower inclinations than those of the peridotitic samples. This directional difference suggests the possibility of a time lag in ChRM acquisition between the basaltic and peridotitic komatiites, while the quantity and quality of the directional data for the basaltic samples are insufficient to settle the matter. The dissimilarities in NRM intensity and in demagnetization behavior between these two types of komatiite are also indicative of different origins of ChRM.

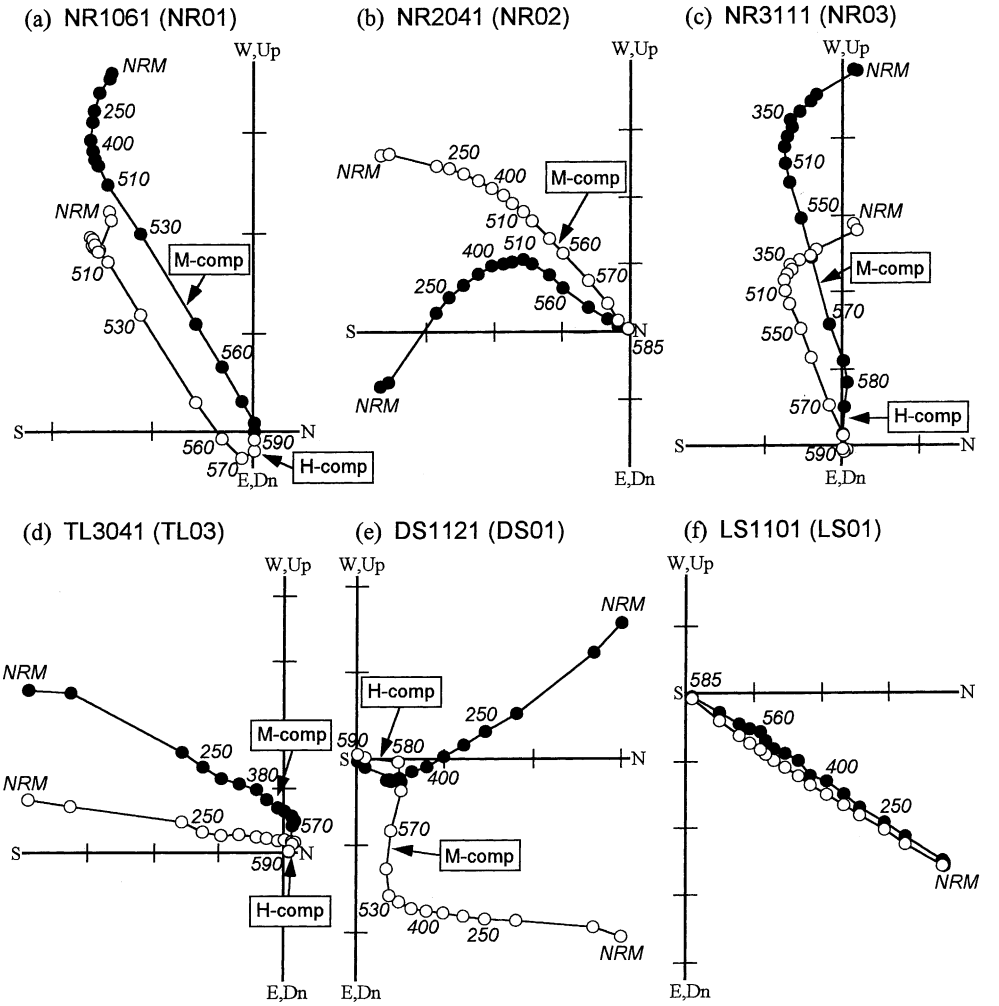


Fig. 8. Results of thermal demagnetization for peridotitic komatiite samples from the Belingwe greenstone belt. For an explanation of the diagrams, see Fig. 6. H-comp: high-temperature component. M-comp: middle-temperature component.

3.2. Results from komatiites of the Belingwe greenstone belt

Generally, komatiite samples from the Reliance Formation are strongly magnetized. Peridotitic samples from sites DS01, DS02, TL03, NR01, NR02, NR03, LS01 and LS02 have large NRM intensity values typically ranging from 1 to 10 A/m. However, magnetization was much weaker in samples from sites TL02 and LS03, being in the order of 10^{-2} to 10^{-1} A/m, and reliable remanences were not obtained from TL01 samples. This probably indicates the composition of the samples from these three sites to be basaltic. Site-mean Q_n ratios for the peridotitic komatiites ranged from 1 to 72, suggesting their stable nature against the presently induced magnetization. As shown in Fig. 5b, the peridotitic and basaltic samples yielded obviously different trends in the Q_n–NRM plot. Table 2 summarizes details of NRM intensities and Q_n ratios.

Three to 11 samples from each site were thermally demagnetized up to 585–600 °C. Because the basaltic samples were too weakly magnetized to demagnetize or were unstable to thermal demagnetization, only peridotitic komatiites were used for the following investigations. Typical thermal demagnetization curves are shown in Fig. 8. The NRM of the peridotitic

samples from sites DS01, NR01, NR02, NR03 and TL03 mainly consists of two or three components (Fig. 8a–e). Low-temperature components that appear to include viscous remanences were removed by heating up to 400–500 °C, and middle-temperature components were isolated at temperatures ranging from 400–500 °C to 570–580 °C. Moreover, in many samples, much smaller high-temperature components can be observed in the narrow range from 570–580 °C to 590–600 °C. On the other hand, the results of thermal demagnetization for samples from sites DS02, LS01 and LS02 were quite different. As shown in Fig. 8f, these samples almost all have completely single-component magnetizations with widely scattered directions even within the same site. Especially, the samples from sites LS01 and LS02 can be characterized by much larger NRM intensities and Q_n ratios than those of the other sites as shown in the Q_n–NRM plot (Fig. 5b), suggesting acquisition of their NRM from a very intense magnetic field. This is most naturally explained as a result of lightning-induced isothermal remanent magnetization (IRM) acquisition. Thus, samples from these three sites were not used for the following investigations.

Middle-temperature and high-temperature components were isolated from 40 and 38 samples, respectively, from five sites. Their site-mean directions were calculated. A tilt correction was also made for the in situ site-mean directions. Site DS01 is located close to the borehole for geological investigations by Renner et al. (1994), in which they reported a value of bedding of the Reliance Formation, thus this value was used for the bedding at site DS01. Values of the strike of the strata at the other sites were estimated from the geological map of the greenstone belt presented by Martin et al. (1993). For the dip at sites NR01, NR02 and NR03, the value measured at an outcrop near site LS03 was used. For site TL03, the value of 75°, which is a typical dip of the Reliance Formation in the sampling area, was applied, since there is no available bedding information around this site. Details of directional data are given in Table 4. Fig. 9a and b show in situ and tilt-corrected site-mean directions, respectively, of the high-temperature components with their 95% confidence limits. Directions were not well clustered within each site, showing considerably large values of α_{95}. Site-mean directions were widely scattered before and after tilt correction. There is no strong evidence to interpret their origin, but one possibility is chemical remanences possessed by very fine-grained magnetites secondarily formed by minor metamorphic events. On the other hand, the middle-temperature component directions were well clustered within each site (Fig. 9c). By applying tilt correction, the site-mean directions were also well grouped (Fig. 9d), enhancing the precision parameter,

Table 4
Paleomagnetic directions of komatiites from the Belingwe greenstone belt

Site	Bedding	n	In situ		Tilt-corrected		α_{95} (°)	k
	Strike/dip (°)		Dec (°)	Inc (°)	Dec (°)	Inc (°)		
High-temperature component								
DS01	135/72	8	10.6	27.5	281.6	57.6	17.4	11.1
NR01	345/82	8	277.4	56.0	59.1	38.8	17.0	11.6
NR02	345/82	5	231.0	29.5	118.3	58.8	61.7	2.5
NR03	345/82	8	276.4	10.1	352.2	68.8	5.2	115.4
TL03	250/75	9	280.2	−5.9	252.0	−30.7	19.7	7.8
Middle-temperature component								
DS01	135/72	7	174.5	74.3	212.9	7.7	6.0	101.4
NR01	345/82	8	238.3	−33.5	235.0	45.5	3.5	246.1
NR02	345/82	7	218.9	−29.1	213.4	39.2	18.1	12.0
NR03	345/82	8	249.0	−41.1	249.1	40.5	3.5	250.2
TL03	250/75	10	210.2	1.0	239.0	38.5	11.8	17.7
Mean(tilt-corrected)		5			228.8	35.3	19.4	16.4

n, number of samples used for paleodirectional determinations; Dec, mean declination; Inc, mean inclination; α_{95}, 95% confidence limit; k, Fisherian precision parameter.

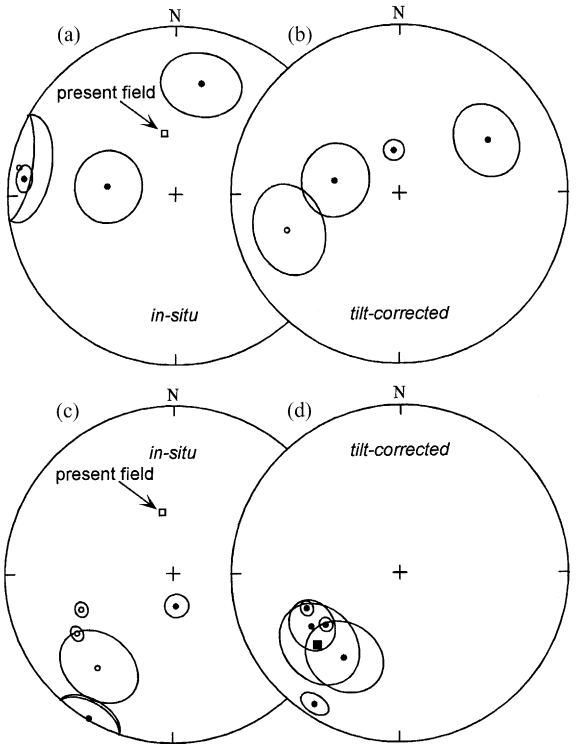


Fig. 9. Mean paleomagnetic directions of komatiite samples from the Belingwe greenstone belt with 95% confidence limits. For an explanation of the projections, see Fig. 7. (a) In situ site-mean directions of the high-temperature components. (b) Tilt-corrected site-mean directions of the high-temperature components. (c) In situ site-mean directions of the middle-temperature components. (d) Tilt-corrected site-mean directions of the middle-temperature components. Tilt-corrected formation-mean of the middle-temperature components is represented by a solid square.

k, from 2.8 to 16.4 ($k_s/k_g = 5.8$). This means that the fold test of McElhinny (1964) is positive at the 95% confidence level. For these komatiites, therefore, the middle-temperature components can be regarded as ChRM acquired prior to folding which is considered to have occurred between 2570 and 2692 Ma (Bickle et al., 1993).

4. Rock magnetic properties

In Section 3, it was suggested from their blocking temperatures that the NRM of the peridotitic komatiites from both greenstone belts was mainly carried by magnetite. More detailed rock magnetic properties were investigated by magnetic hysteresis experiments and thermomagnetic analyses with a Princeton vibrating sample magnetometer. The characteristics of the magnetic minerals in the komatiite samples from both greenstone belts were found to be very similar.

Hysteresis loop measurements were made for 25 peridotitic samples for each of the komatiite formations. Typical loop examples are shown in Fig. 10, and hysteresis parameters are listed in Tables 5 and 6. When two or three specimens taken from a single sample were measured, indices a, b, and c were appended onto the sample name as specimen codes in the tables. One can see that, in some cases (e.g., BT601), there are considerable differences in hysteresis parameters between two specimens taken from the same sample. This might reflect nonuniform spatial distribution of magnetic minerals with certain grain sizes (see Section 6). All of the samples from both greenstone belts yielded coercivity (H_c) values less then 20 mT, except that the only one sample from Barberton (BT205) showed a much higher value of 44.7 mT. This low-coercivity nature is also indicative that the main magnetic mineral in the samples is magnetite. Magnetite content derived from saturation magnetization (M_s) values was 2.2–6.5 wt.% for Barberton and 0.2–3.4 wt.% for Belingwe. To obtain insight into the distribution of the magnetic domain state of the samples, hysteresis loop parameters, M_{rs}/M_s and H_{cr}/H_c (M_{rs}: saturation remanent magnetization, H_{cr}: coercivity of remanence) were plotted on Day plots (Day et al., 1976, 1977) (Fig. 11). One can see that the data for both greenstone belts are generally compatible with pseudo single domain (PSD) trends. In magnetite, however, PSD trends are virtually indistinguishable from single domain (SD)–multidomain (MD) mixed curves (Dunlop, 2002). As will be revealed by microscopic observations (see Section 6), it seems more natural to interpret these observed trends as SD + MD mixed curves.

Thermomagnetic analyses were performed with six samples for each greenstone belt. The samples were heated by helium gas flow, but air was not excluded from the furnace. Typical results are shown in Fig. 12. Curie temperatures in the Barberton samples ranged from 586 to 595 °C, and those from the Belingwe samples ranged from 589 to 597 °C, indicating that the main magnetic mineral is Ti-free pure magnetite.

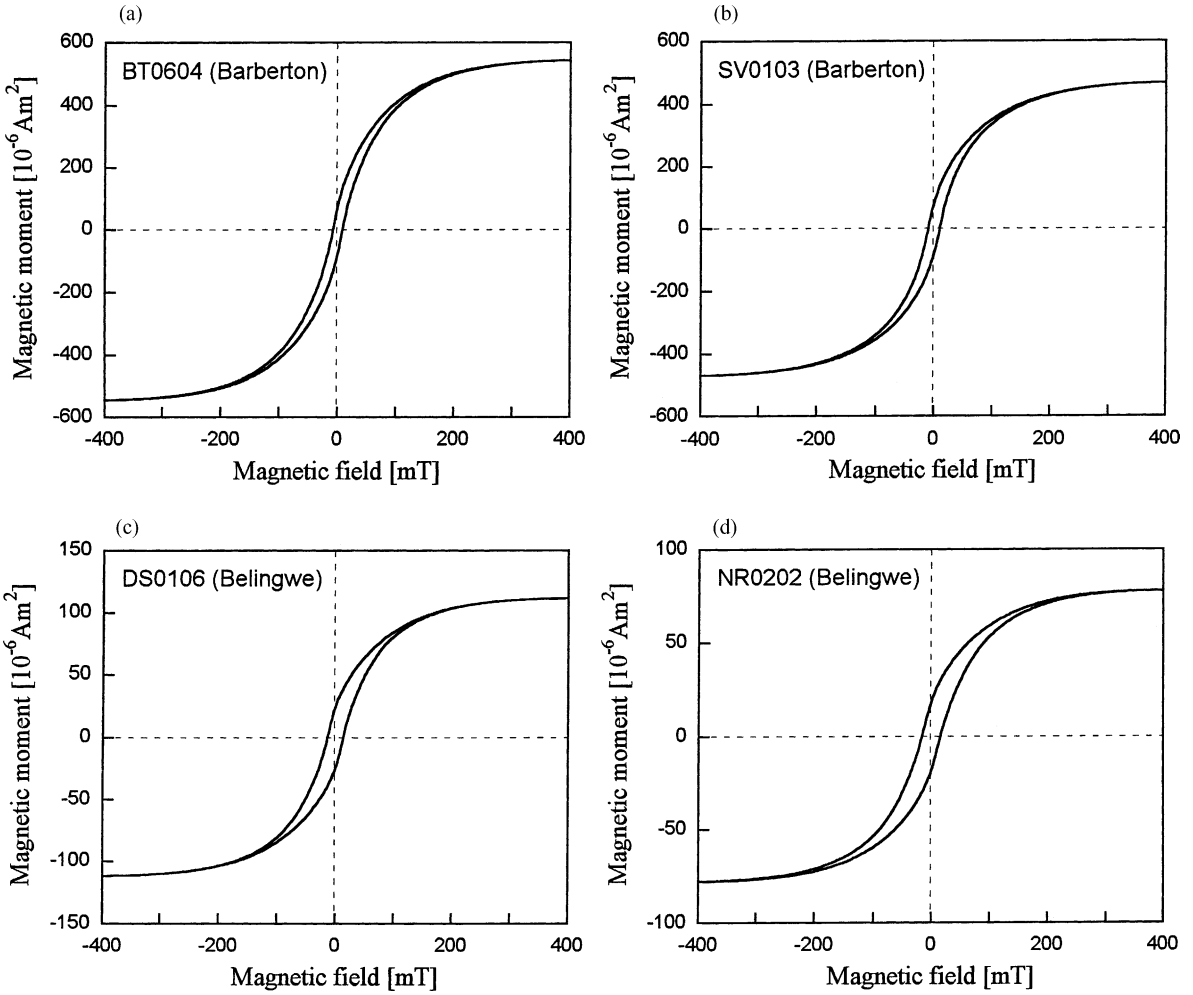


Fig. 10. Hysteresis loops for samples from the Barberton (a, b) and Belingwe (c, d) greenstone belts. Paramagnetic components were subtracted.

The reversible nature of the thermomagnetic curves is indicative of the minimal effect on their magnetic properties by heating up to 650–700 °C. The degree of chemical alterations by heating was examined more minutely with two samples for each greenstone belt by hysteresis measurements after stepwise heating. Fig. 13 shows hysteresis loops at room temperature before and after heating up to 640 °C for a Barberton sample (Fig. 13a) and 700 °C for a Belingwe sample (Fig. 13b), along with graphs of the change in normalized M_s at room temperature by stepwise heating (right insets). This figure shows that the loops after heating to 640–700 °C remained almost unchanged from those before heating, with virtually constant M_s values after stepwise heating. This also indicates that the magnetic properties of the komatiite samples from both greenstone belts are conspicuously stable when subjected to laboratory heating up to 600–700 °C.

5. Paleointensity measurements

5.1. Experimental details

Paleointensity measurements by Coe's (1967) modified version of the Thellier–Thellier method

Table 5
Hysteresis parameters for komatiites from the Barberton greenstone belt

Site	Sample	M_s (A m^2/kg)	M_r (A m^2/kg)	H_c (mT)	H_{cr} (mT)	M_r/M_s	H_{cr}/H_c
BT02	BT202	5.21	0.63	6.6	11.3	0.12	1.70
	BT204a	4.67	0.79	8.0	11.8	0.17	1.47
	BT204b	4.47	0.78	8.6	13.4	0.17	1.55
	BT205	2.50	1.31	44.7	59.4	0.53	1.33
BT06	BT601a	3.46	0.89	19.2	43.2	0.26	2.25
	BT601b	3.71	0.71	10.2	20.9	0.19	2.06
	BT604	4.33	0.62	8.0	18.5	0.14	2.32
	BT606	3.98	0.58	8.0	14.7	0.15	1.85
BT07	BT702	4.47	0.62	5.6	8.8	0.14	1.57
	BT703	5.51	0.37	6.6	20.9	0.07	3.15
	BT704	4.39	0.60	5.7	9.7	0.14	1.69
BT09	BT902a	5.18	0.27	3.8	9.3	0.05	2.45
	BT902b	5.27	0.25	3.8	9.7	0.05	2.55
	BT906	5.62	0.36	3.8	7.4	0.06	1.97
	BT907	6.02	0.43	4.6	11.8	0.07	2.57
SV01	SV101	4.45	0.88	9.9	15.9	0.20	1.61
	SV103	3.85	0.67	9.8	17.4	0.17	1.77
	SV105	4.61	0.49	6.0	11.2	0.11	1.87
	SV108a	4.75	1.02	12.8	21.4	0.21	1.68
	SV108b	5.15	1.03	12.5	17.0	0.20	1.36
SV02	SV201	3.20	0.78	11.8	17.5	0.24	1.48
	SV203	3.43	0.83	11.1	15.8	0.24	1.42
	SV207	3.58	0.82	11.4	16.8	0.23	1.48
SV04	SV404	3.51	0.92	12.4	17.6	0.26	1.42
	SV406	4.41	0.91	12.7	20.9	0.21	1.65
	SV407a	3.86	0.60	9.3	15.8	0.15	1.70
	SV407b	3.86	0.86	10.8	16.7	0.22	1.55
TV01	TV101	3.43	0.37	5.5	10.7	0.11	1.92
	TV106	3.19	0.46	13.3	38.3	0.14	2.88
	TV112a	2.14	0.40	15.6	46.1	0.19	2.96
	TV112b	2.06	0.36	14.8	39.8	0.17	2.68

M_s, saturation magnetization; M_r, saturation remanent magnetization; H_c, coercivity; H_{cr}, coercivity of remanence.

(Thellier and Thellier, 1959) were performed on peridotitic komatiite samples from both greenstone belts from which the ChRM components were obtained. All heat treatments were undertaken in air with the above-mentioned thermal demagnetizer, and the magnetizing field applied was controlled at 15 and 10 μT for the Barberton and Belingwe samples, respectively. Double-heating procedures were repeated at incrementally higher temperatures in more than 11 steps, and after each heat treatment, initial magnetic susceptibility at room temperature was monitored to detect chemical alterations. In addition, the pTRM test (Coe, 1967), which checks reproducibility of pTRM, was performed several times during the double-heating sequences. Remaining and acquired remanences after the double-heating processes were plotted on the NRM–TRM diagram, in which a least-squares fitting was made for data points corresponding to the ChRM component with the treatment according to York (1966, 1967). To evaluate the reliability of the paleointensity determinations, the factors *f*, *g* and *q* relating to quality of the data were calculated following the algorithm given by Coe et al. (1978). Criteria for data analysis were based on the following three points. (1) The linear segment in the NRM–TRM diagram should consist of more than four data points

Table 6
Hysteresis parameters for komatiites from the Belingwe greenstone belt

Site	Sample	M_s (A m^2/kg)	M_r (A m^2/kg)	H_c (mT)	H_{cr} (mT)	M_r/M_s	H_{cr}/H_c
DS01	DS105	1.33	0.24	12.8	24.1	0.18	1.88
	DS106	0.88	0.20	13.3	23.1	0.22	1.74
	DS112a	0.64	0.15	14.9	26.5	0.24	1.77
	DS112b	0.89	0.18	11.4	20.2	0.20	1.76
	DS112c	0.83	0.17	13.0	23.0	0.21	1.77
	DS201	2.82	0.37	7.2	14.8	0.13	2.06
DS02	DS204	2.88	0.41	7.8	16.3	0.14	2.09
	DS206a	2.25	0.31	6.5	12.0	0.14	1.85
	DS206b	2.02	0.29	7.0	14.2	0.14	2.04
LS01	LS106	1.15	0.15	7.3	15.9	0.13	2.19
	LS108	0.56	0.10	10.9	18.5	0.19	1.70
	LS110	0.57	0.12	7.4	12.2	0.21	1.65
	LS112a	0.44	0.11	14.2	19.8	0.24	1.40
	LS112b	0.48	0.12	9.9	16.7	0.25	1.69
LS02	LS201	0.38	0.10	8.9	13.0	0.27	1.46
	LS207	0.48	0.11	10.9	20.4	0.23	1.87
	LS210a	1.19	0.13	8.1	17.9	0.11	2.22
	LS210b	1.08	0.12	9.1	20.0	0.12	2.20
TL03	TL303	2.50	0.20	4.9	11.3	0.08	2.29
	TL305a	2.23	0.17	4.9	11.0	0.07	2.25
	TL305b	2.31	0.17	4.8	10.8	0.07	2.23
	TL309	3.17	0.28	5.3	11.3	0.09	2.12
NR01	NR101	0.48	0.12	12.6	21.6	0.25	1.71
	NR103	0.28	0.09	18.4	28.5	0.33	1.55
	NR108a	0.20	0.05	14.6	25.7	0.27	1.77
	NR108b	0.93	0.13	8.2	20.3	0.14	2.47
NR02	NR202	0.56	0.13	15.7	33.6	0.24	2.14
	NR204	1.12	0.23	13.9	25.2	0.21	1.81
	NR211a	0.62	0.13	10.4	17.5	0.21	1.68
	NR211b	0.72	0.15	10.3	17.7	0.21	1.73
NR03	NR304	1.32	0.21	10.6	22.3	0.16	2.10
	NR309	0.87	0.23	18.9	39.7	0.26	2.10
	NR310	1.01	0.21	10.8	20.1	0.21	1.87

M_s, saturation magnetization; M_r, saturation remanent magnetization; H_c, coercivity; H_{cr}, coercivity of remanence.

and (2) should be selected to make the quality factor, *q*, maximum within the temperature range in which the correlation coefficient of the data points exceeds 0.98. (3) The results of the pTRM test should be reproduced within the experimental errors of previously acquired TRM in the temperature range of the linear segment. For calculations of experimental errors with the method proposed by Kono and Tanaka (1984), we used error estimates from various sources, such as errors in applied magnetic field controls of 0.1 μT, in intensity measurements of 2.5%, in direction measurements of 2.5°, etc.

5.2. Results from komatiites of the Barberton greenstone belt

Generally, the peridotitic komatiite samples were very stable when subjected to double-heating procedures in Thellier–Thellier experiments, and most samples yielded successful results. Fig. 14 shows successful examples (that passed above criteria) of NRM–TRM diagrams with graphs of changes in normalized susceptibility and thermal demagnetization curves. One can see that the temperature range of the linear segment to calculate the slope corre-

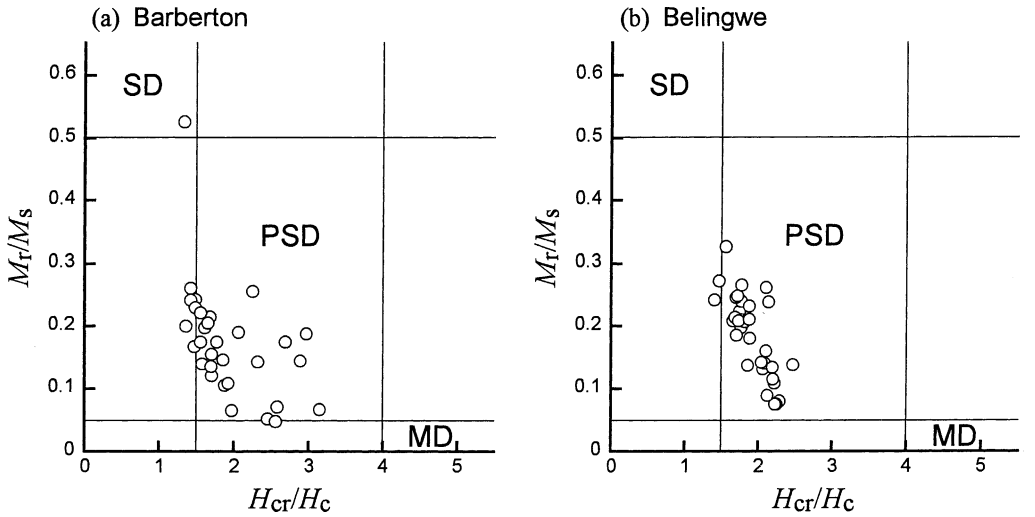


Fig. 11. Day plots (Day et al., 1976, 1977) for samples from the Barberton (a) and Belingwe (b) greenstone belts. SD: single domain. PSD: pseudo single domain. MD: multidomain.

sponds to the high-temperature component that can be isolated in the demagnetization curve. In these examples, the pTRM tests are well satisfied within the linear segments consisting of 5–9 data points whose correlation coefficients exceed 0.99. Susceptibility changes observed in the temperature ranges of the linear segments for results that successfully met the criteria were typically less than 20% of the initial values.

Paleointensity determinations were successful for 28 peridotitic samples from eight sites. Table 7 summarizes these results. All of the successful samples yielded considerably small paleointensity values. Following the algorithm given by Coe et al. (1978), we calculated a weighted site-mean paleointensity for each site using the quality factor, q, as a weighting factor. This might not represent a cooling unit-mean but probably a mean value for two or three lava flows. The weighted site-mean intensities obtained from eight sites ranged from $2.7 \pm 0.1\ \mu T$ to $18.7 \pm 3.3\ \mu T$. A corresponding range of virtual dipole moments (VDMs) calculated with the site-mean inclinations was from $(0.7 \pm 0.2) \times 10^{22}\ A\,m^2$ to $(4.8 \pm 0.9) \times 10^{22}\ A\,m^2$. The VDM for site TV01 was calculated with the tilt-corrected formation-mean inclination. Averaging the VDMs for eight sites yielded a very weak value of $(1.8 \pm 1.3) \times 10^{22}\ A\,m^2$, which corresponds to 24% of the present dipole moment.

5.3. Results from komatiites of the Belingwe greenstone belt

More than half of the samples from sites DS01, NR01, NR02 and NR03 yielded successful results. Fig. 15 shows typical examples of NRM–TRM diagrams with graphs showing changes in normalized susceptibility and thermal demagnetization curves. The temperature ranges of the linear segments were within the middle-temperature components corresponding to the ChRM, as can be seen in the demagnetization curves. In these examples, the pTRM tests were well satisfied within linear segments consisting of four to seven data points whose correlation coefficients exceed 0.98. Susceptibility changes observed in the temperature ranges of the linear segments for successful results were typically less than 20% of initial values.

Paleointensity determinations were successful for 29 samples from four sites. Table 8 summarizes the results. As in the case of komatiites from Barberton, all of the successful results showed very weak intensity values. Weighted site-mean paleointensities for four sites ranged from $1.1 \pm 0.4\ \mu T$ to $12.4 \pm 1.0\ \mu T$. A corresponding range of VDMs calculated with the site-mean inclinations was from $(0.3 \pm 0.1) \times 10^{22}\ A\,m^2$ to $(2.5 \pm 0.2) \times 10^{22}\ A\,m^2$. Averaging the values from four sites yielded a very weak value of

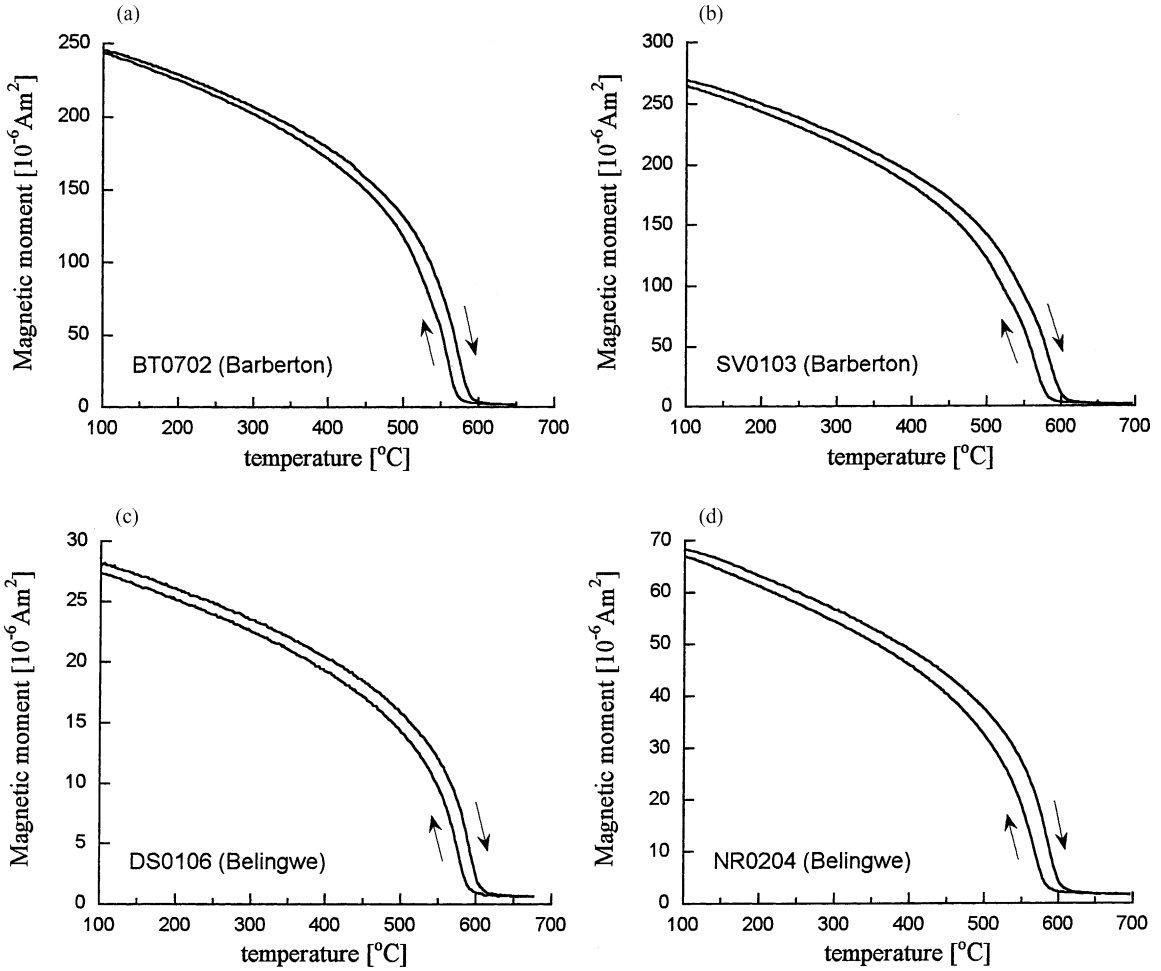


Fig. 12. Results of thermomagnetic analyses of samples from the Barberton (a, b) and Belingwe (c, d) greenstone belts.

$(1.1 \pm 0.9) \times 10^{22}$ A m^2, which corresponds to 15% of the present dipole moment.

6. Microscopic observations

As described in Section 4, the NRM of the peridotitic komatiites from the Barberton and Belingwe greenstone belts is mainly carried by pure magnetite, and the rock magnetic properties of the komatiites from the two greenstone belts are very similar. It is well known that these magnetites were formed during serpentinization as secondary minerals (Viljoen and Viljoen, 1969b; Bickle et al., 1993).

Mineralogy of the peridotitic komatiite samples from each greenstone belt was investigated by a scanning electron microscope (SEM), and the composition of the minerals was analyzed by an energy dispersive X-ray analyzer (EDX) fitted with the SEM. Fig. 16a and b show electron micrographs of a sample from Belingwe, and Fig. 16c and d are those of a Barberton sample. The mineral assemblage and texture of the samples from both greenstone belts are very similar. In Fig. 16a and c, many dark gray phenocrysts, typically 100–400 μm in size, which were originally olivine phenocrysts are evident. Within the phenocryst, the composition of the core (light area) is represented by olivine, indicating a very high MgO content, while

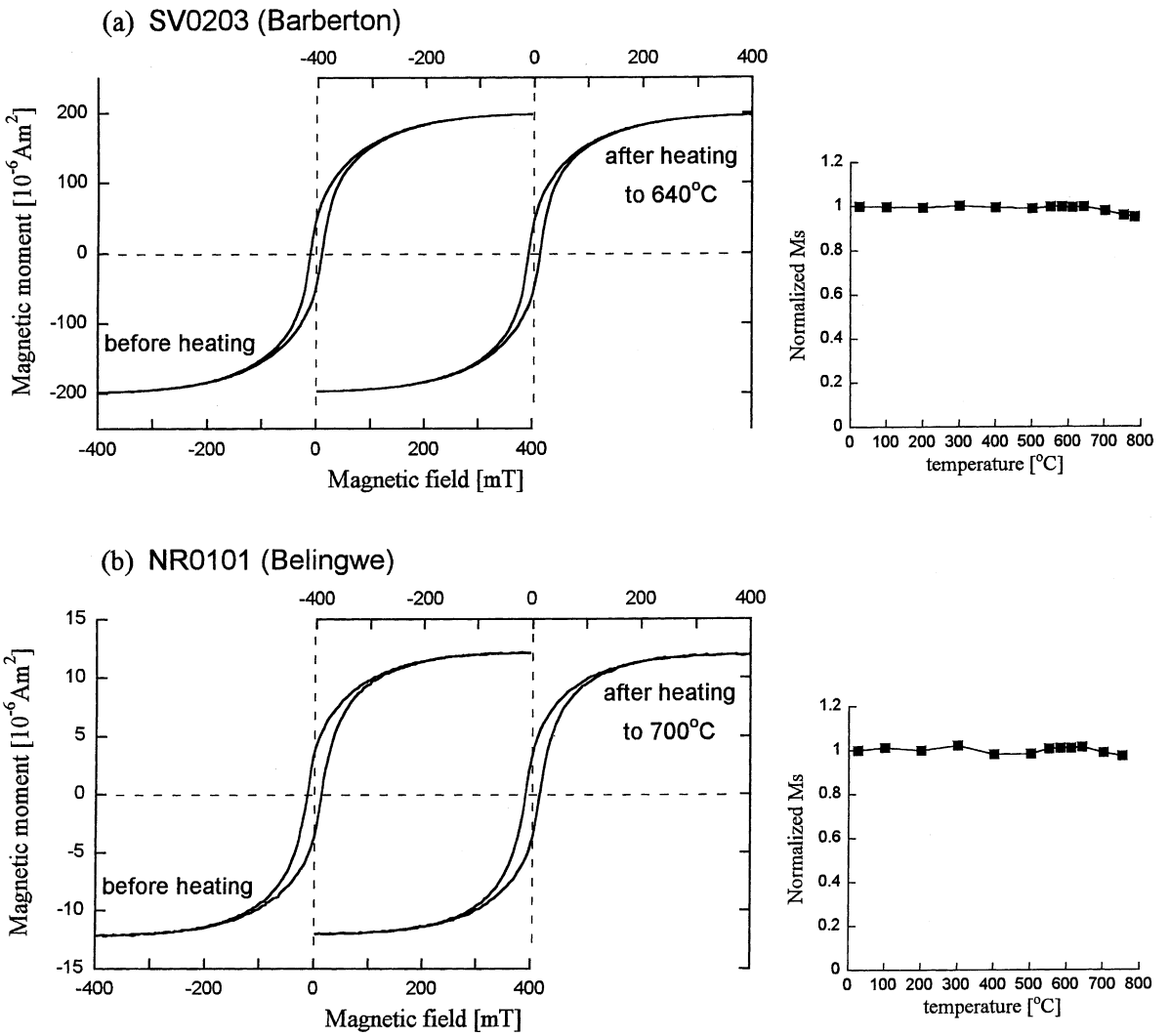


Fig. 13. Hysteresis loops measured at room temperature before and after heating to 640–700 °C of samples from the Barberton (a) and Belingwe (b) greenstone belts, with graphs of the change in normalized M_s at room temperature by stepwise heating (right insets). Paramagnetic components were subtracted.

the outer part (dark area) in the phenocryst has been completely serpentinized surrounding the olivine core. Generally, these olivine-serpentine phenocrysts have a very low Fe content. Much smaller grains and veinlets of pure magnetite (white area) form the outer rims of the olivine-serpentine phenocrysts, which are obviously products of the serpentinization of olivine. While the veinlet-shaped magnetites are large, in the order of 10–100 μm, the size of the grain-shaped magnetite varies, typically ranging from <1 to 10 μm, as can be seen in Fig. 16b and d. A much larger white crystal with an euhedral shape is original igneous chrome-spinel (Fig. 16a and c), in which a considerably high Fe content can be detected, as well as a high chrome content. Cr–Fe spinels are seldom major contributors to remanences (Dunlop and Özdemir, 1997). The existence of the magmatic chrome-spinels indicates that titanomag-

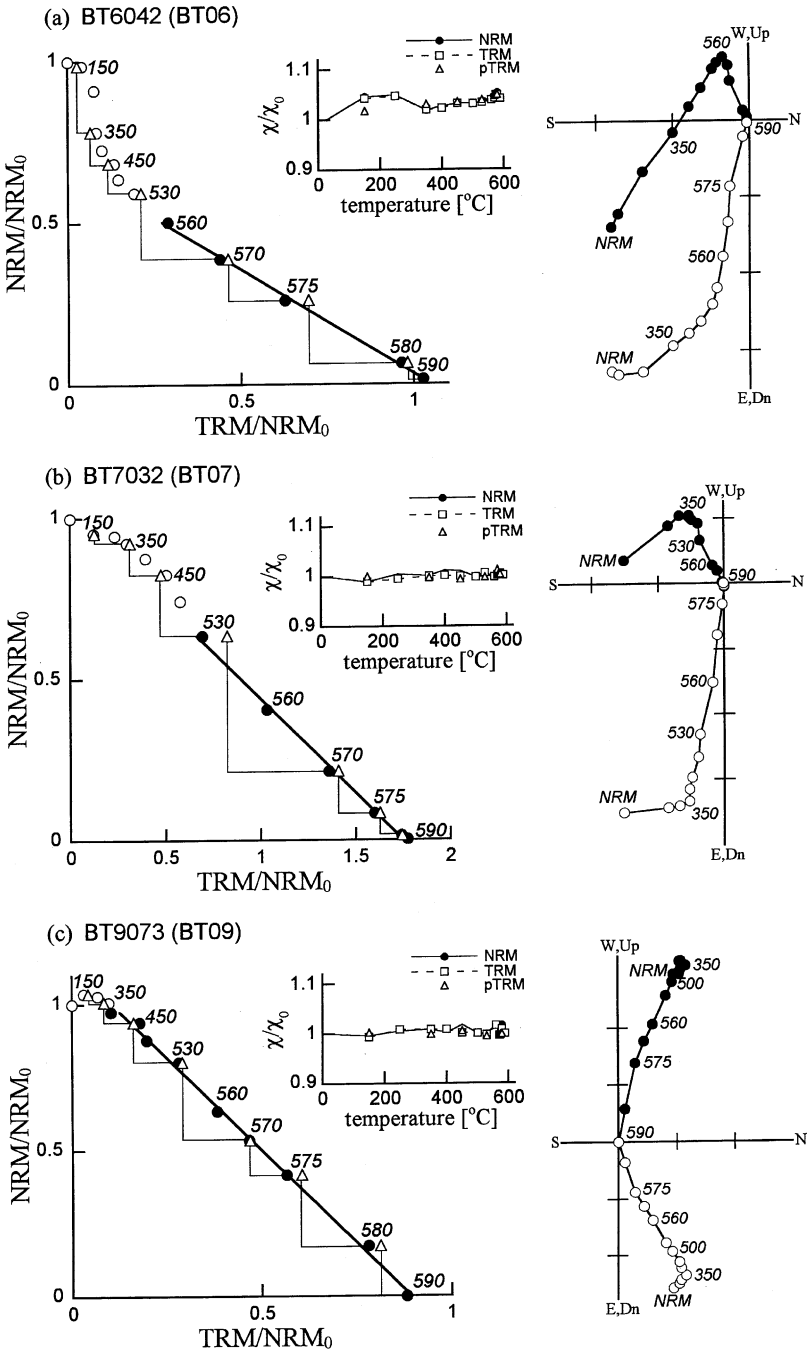


Fig. 14. Typical successful results of Thellier–Thellier experiments for komatiite samples from the Barberton greenstone belt. NRM–TRM diagrams (left) with results of susceptibility measurements (middle-upper) and demagnetization curves (right). In the NRM–TRM diagram, units of the NRM and TRM axes are normalized by the original NRM intensity (NRM_0). Solid (open) circles indicate the data points used (rejected) for straight line fitting. Numbers beside the points represent temperatures (°C). Open triangles show data points of the pTRM test. Susceptibility values (χ) are normalized by the original value (χ_0).

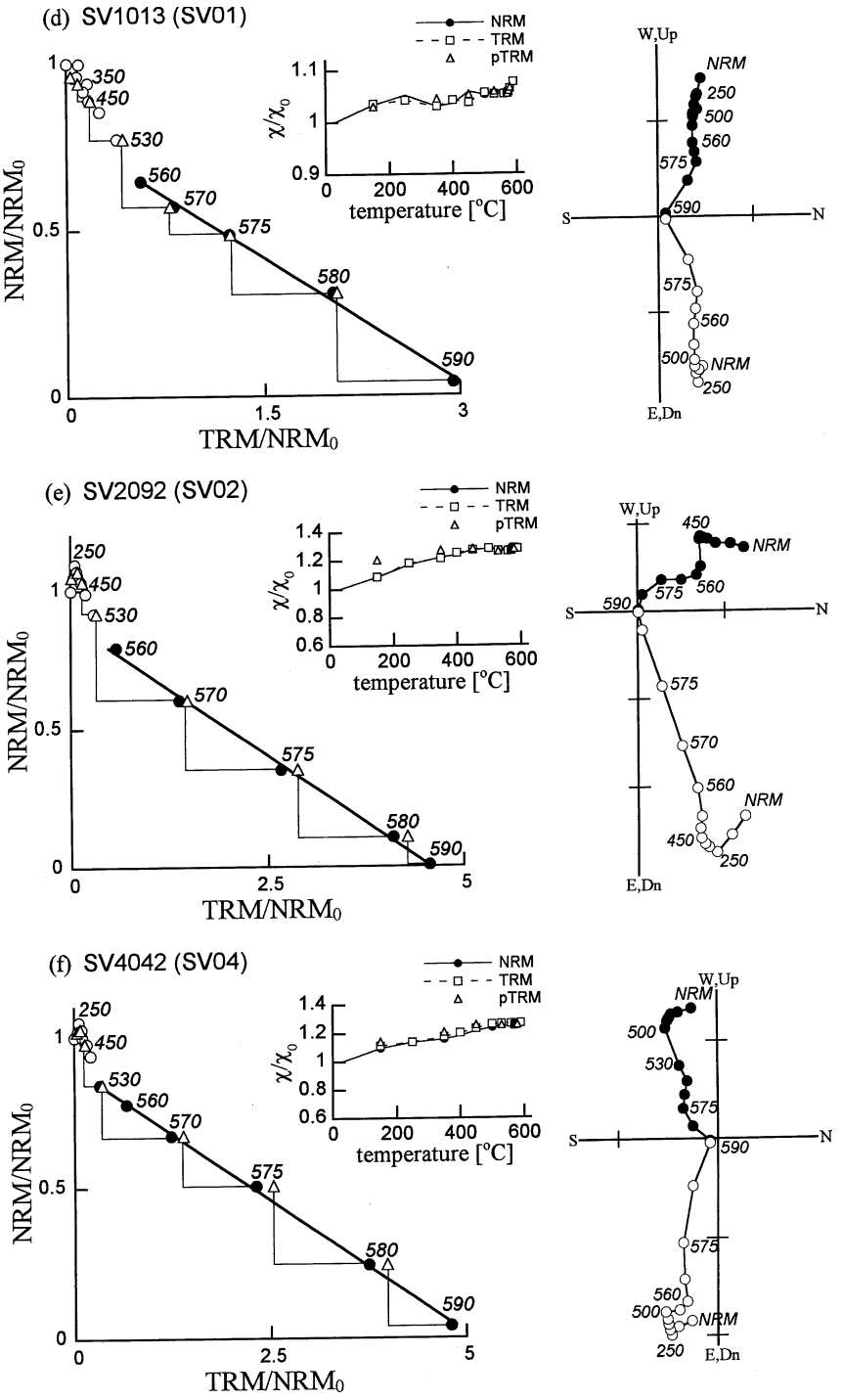


Fig. 14. (*Continued*).

Table 7
Results of Thellier–Thellier experiments for komatiites from the Barberton greenstone belt

Site	Sample	ΔT (°C)	n	r	f	g	q	$F \pm \sigma$ (μT)	$F_m \pm \sigma_m$ (μT)	VDM $\pm \sigma_V$ (10^{22} A m^2)
BT02	BT2022	570–590	4	0.996	0.77	0.61	7.4	12.6 ± 0.8	11.0 ± 1.2	2.77 ± 0.30
	BT2043	560–590	5	0.999	0.84	0.67	19.5	10.8 ± 0.3		
BT06	BT6012	560–590	5	0.999	0.78	0.69	30.1	7.6 ± 0.1	5.9 ± 1.0	1.52 ± 0.26
	BT6022	560–590	5	0.999	0.54	0.69	20.9	4.4 ± 0.1		
	BT6042	560–590	5	0.999	0.71	0.69	16.8	9.7 ± 0.3		
	BT6052	560–590	5	0.999	0.67	0.71	21.8	5.2 ± 0.1		
	BT6062	560–590	5	0.999	0.73	0.71	24.4	6.2 ± 0.1		
BT07	BT7022	570–590	4	0.999	0.68	0.63	11.1	7.7 ± 0.3	8.8 ± 1.4	2.26 ± 0.36
	BT7032	530–590	6	0.998	0.62	0.73	15.1	8.7 ± 0.3		
	BT7042	570–590	4	0.998	0.73	0.62	9.9	12.0 ± 0.5		
BT09	BT9022	400–590	9	0.999	0.86	0.84	56.9	20.8 ± 0.3	18.7 ± 3.3	4.83 ± 0.85
	BT9063	350–590	10	0.992	0.85	0.85	15.8	10.9 ± 0.5		
	BT9073	400–590	9	0.998	0.86	0.84	30.9	19.0 ± 0.4		
SV01	SV1013	560–590	5	0.998	0.76	0.69	14.2	3.8 ± 0.1	3.9 ± 0.5	0.92 ± 0.12
	SV1022	560–580	4	0.985	0.41	0.61	2.0	3.4 ± 0.4		
	SV1032	560–580	4	0.983	0.33	0.59	1.5	4.2 ± 0.5		
	SV1051	560–580	4	0.988	0.40	0.54	2.0	6.0 ± 0.6		
	SV1062	570–590	4	0.999	0.78	0.62	13.0	4.1 ± 0.2		
SV02	SV2042	560–580	4	0.998	0.46	0.58	5.3	2.3 ± 0.1	2.7 ± 0.1	0.68 ± 0.03
	SV2051	570–590	4	0.999	0.71	0.65	20.3	2.7 ± 0.1		
	SV2062	570–590	4	0.999	0.76	0.64	17.3	2.4 ± 0.1		
	SV2092	560–590	5	0.999	0.88	0.72	21.2	2.9 ± 0.1		
SV04	SV4022	450–590	8	0.999	0.84	0.83	34.7	6.4 ± 0.1	2.9 ± 1.0	0.66 ± 0.23
	SV4042	530–590	6	0.999	0.89	0.76	44.5	2.7 ± 0.1		
	SV4062	570–590	4	0.999	0.72	0.66	36.4	2.5 ± 0.1		
	SV4073	570–590	4	0.999	0.68	0.66	31.7	3.7 ± 0.1		
TV01	TV1062	530–580	5	0.989	0.65	0.72	5.4	4.2 ± 0.1	4.3 ± 0.1	1.08 ± 0.03[a]
	TV1122	530–580	5	0.999	0.67	0.68	44.1	4.3 ± 0.1		

ΔT, temperature range of the linear segment; n, number of data points in the linear segment; r, correlation coefficient of the linear relationship; f, fraction of the total extrapolated NRM spanned by the linear segment; g, gap factor for data points in the linear segment; q, quality factor of the paleointensity estimate; $F \pm \sigma$, paleointensity obtained from each sample with its standard error; $F_m \pm \sigma_m$, weighted site-mean paleointensty with its standard error; VDM $\pm \sigma_V$, virtual dipole moment with its standard error.

[a] VDM for TV01 was calculated with the formation-mean inclination.

netite hardly crystallizes as a primary Fe–Ti oxide. The presence of magnetites of various sizes means that they are mixture of SD and MD grains. Thus, it is more natural to interpret the trends in the Day plots for these samples (see Section 4) not as PSD trends but as SD + MD mixed curves. The ChRM of the samples probably is principally carried by SD grains or very small grains akin to SD, since their blocking temperature distributions are concentrated at very high temperature ranges as described in Section 3.

7. Discussion

7.1. Origin of NRM of komatiites

Our SEM observations support the concept that the magnetites in the komatiite samples were formed during serpentinization as secondary minerals. Generally, the formation of a new magnetic mineral in the presence of a magnetic field results in acquisition of CRM. In particular, in the case of magnetite formed by serpentinization, its main magnetization component at

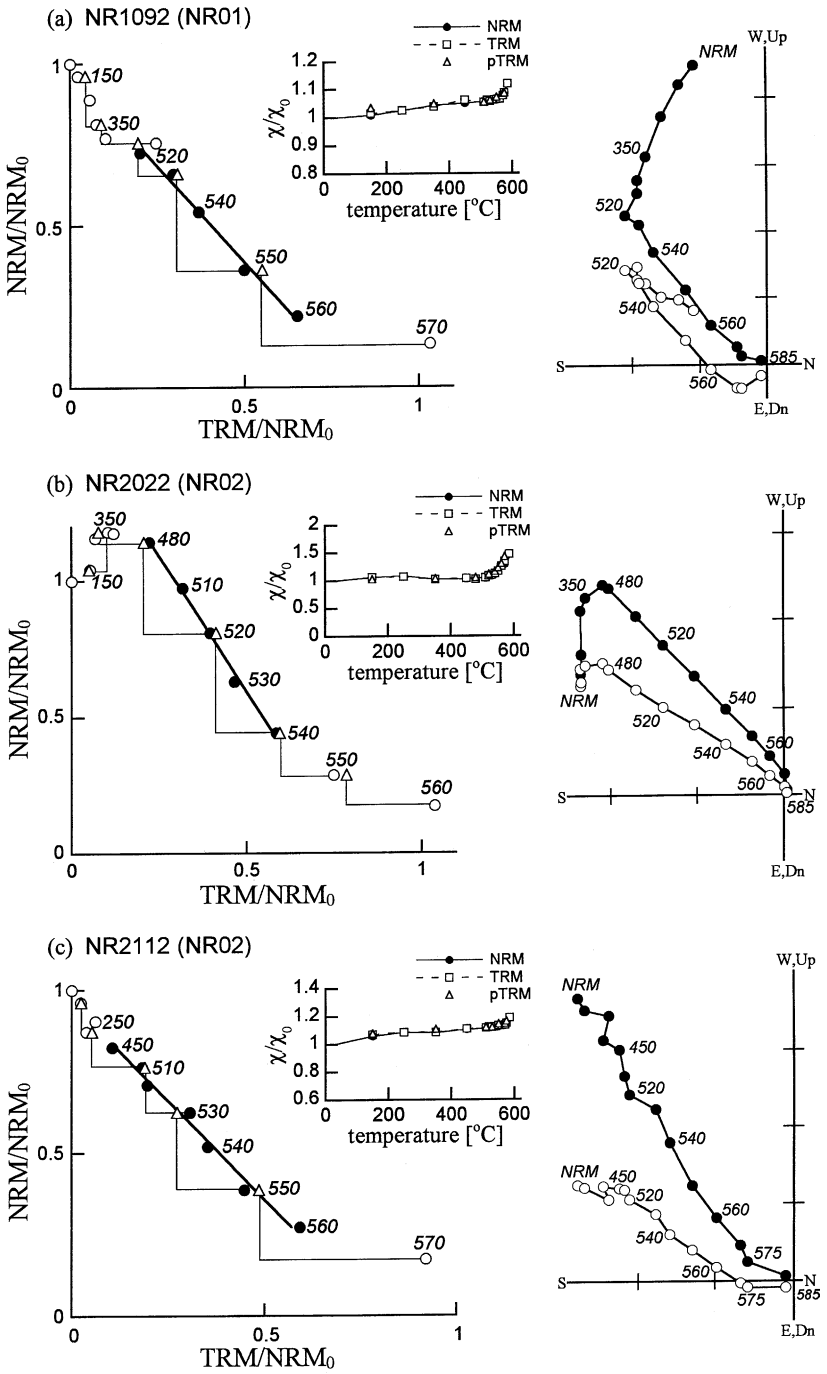


Fig. 15. Typical successful results of Thellier–Thellier experiments for komatiite samples from the Belingwe greenstone belt. For an explanation of the graphs, see Fig. 14.

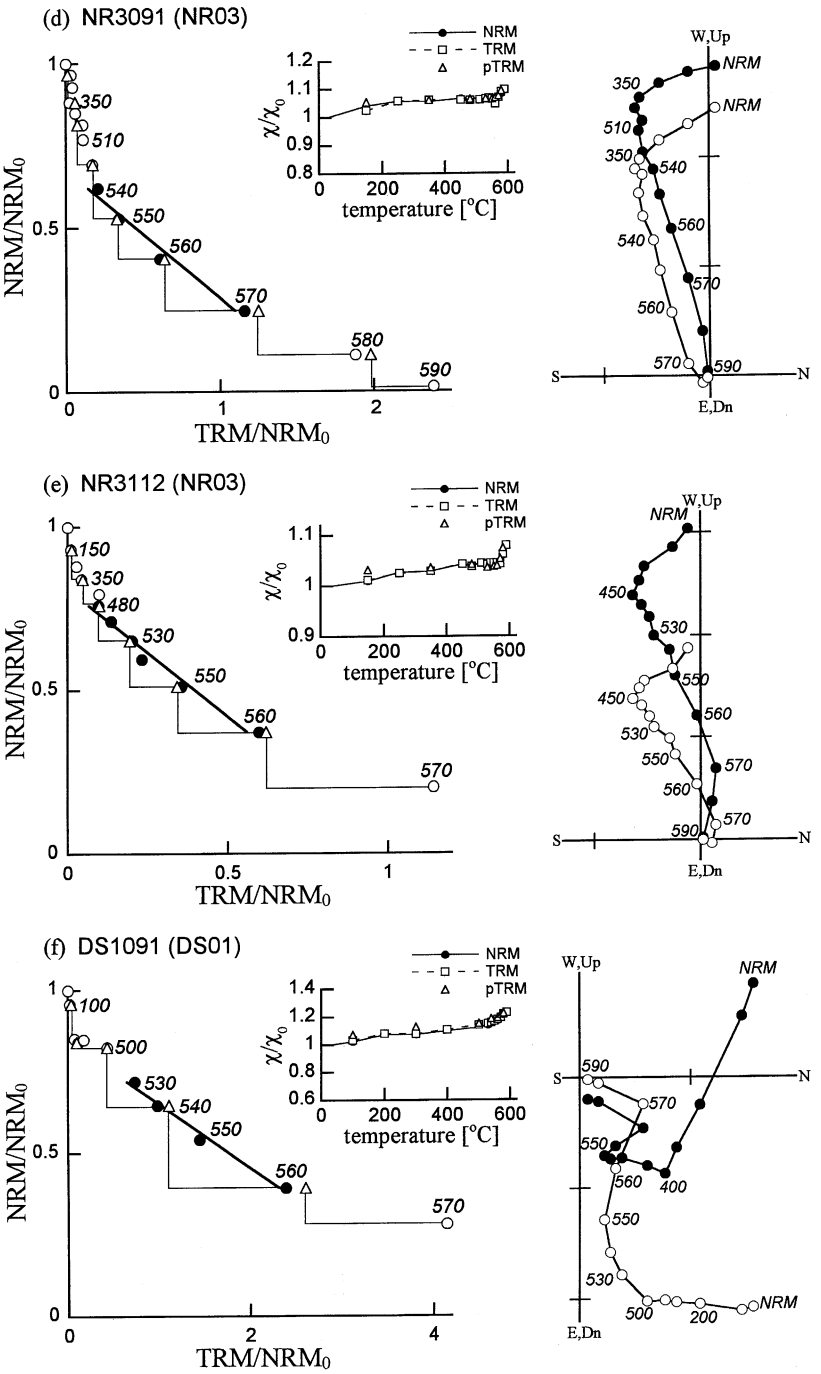


Fig. 15. (*Continued*).

Table 8
Results of Thellier–Thellier experiments for komatiites from the Belingwe greenstone belt

Site	Sample	ΔT (°C)	n	r	f	g	q	$F \pm \sigma$ (μT)	$F_m \pm \sigma_m$ (μT)	VDM $\pm \sigma_V$ (10^{22} A m^2)
NR01	NR1012	520–550	4	0.991	0.36	0.64	2.4	16.4 ± 1.6	12.4 ± 1.0	2.52 ± 0.20
	NR1022	520–550	4	0.984	0.30	0.64	1.5	11.8 ± 1.5		
	NR1032	520–550	4	0.988	0.36	0.63	2.1	11.8 ± 1.3		
	NR1041	510–540	4	0.997	0.30	0.65	3.4	15.7 ± 0.9		
	NR1062	510–550	5	0.985	0.31	0.70	2.2	15.7 ± 1.6		
	NR1072	510–550	5	0.989	0.45	0.69	3.6	11.5 ± 1.0		
	NR1092	520–560	5	0.994	0.53	0.73	6.3	11.8 ± 0.7		
NR02	NR2011	510–540	4	0.999	0.34	0.64	8.3	17.3 ± 0.5	3.0 ± 3.1	0.65 ± 0.67
	NR2022	480–540	5	0.997	0.45	0.75	7.7	19.8 ± 0.9		
	NR2042	530–570	5	0.998	0.62	0.69	12.8	2.6 ± 0.1		
	NR2051	530–560	4	0.998	0.37	0.66	5.0	7.9 ± 0.4		
	NR2061	530–570	5	0.997	0.64	0.64	9.7	2.7 ± 0.1		
	NR2071	530–560	4	0.982	0.24	0.64	1.1	5.2 ± 0.7		
	NR2102	520–550	4	0.995	0.28	0.64	2.6	9.8 ± 0.7		
	NR2112	450–560	7	0.992	0.59	0.81	8.6	12.0 ± 0.7		
	NR2122	510–550	5	0.992	0.43	0.71	4.3	11.3 ± 0.8		
NR03	NR3022	520–560	5	0.996	0.20	0.69	2.8	8.9 ± 0.4	5.1 ± 1.7	1.09 ± 0.36
	NR3032	550–575	4	0.984	0.50	0.60	2.4	3.1 ± 0.4		
	NR3042	540–570	4	0.980	0.50	0.56	2.0	4.2 ± 0.6		
	NR3082	510–550	4	0.982	0.21	0.61	1.0	13.5 ± 1.8		
	NR3091	540–570	4	0.984	0.55	0.62	2.7	3.9 ± 0.5		
	NR3102	480–550	5	0.996	0.29	0.72	4.3	9.7 ± 0.5		
	NR3112	480–560	6	0.985	0.48	0.73	4.0	7.7 ± 0.7		
DS01	DS1032	530–560	4	0.982	0.17	0.61	0.8	2.4 ± 0.3	1.1 ± 0.4	0.28 ± 0.10
	DS1052	530–560	4	0.999	0.28	0.47	3.9	1.0 ± 0.1		
	DS1062	500–550	4	0.995	0.20	0.65	1.9	2.7 ± 0.2		
	DS1071	530–560	4	0.993	0.33	0.57	2.2	1.5 ± 0.1		
	DS1091	530–560	4	0.992	0.38	0.61	2.6	1.9 ± 0.2		
	DS1123	540–570	4	0.982	0.63	0.56	2.6	1.0 ± 0.1		

ΔT, temperature range of the linear segment; n, number of data points in the linear segment; r, correlation coefficient of the linear relationship; f, fraction of the total extrapolated NRM spanned by the linear segment; g, gap factor for data points in the linear segment; q, quality factor of the paleointensity estimate; $F \pm \sigma$, paleointensity obtained from each sample with its standard error; $F_m \pm \sigma_m$, weighted site-mean paleointensty with its standard error; VDM $\pm \sigma_V$, virtual dipole moment with its standard error.

the first stage of formation is not TRM or thermochemical remanent magnetization (TCRM), but grain growth CRM, because serpentinization occurs only at low temperatures. The ChRM of komatiites from the Belingwe greenstone belt was proved to be pre-folding magnetization by the large-scale fold test. Therefore, the acquisition of CRM can be constrained between ca. 2.6 and 2.7 Ga. In the case of komatiites from the Barberton greenstone belt, however, there is no paleomagnetic evidence for timing of the ChRM acquisition. In general, serpentinization in ultramafic volcanic rocks is attributable to the hydrothermal alteration of olivine or pyroxene. The pseudomorphic textures of these komatiites show that serpentinization has been restricted to the rim of the original olivine phenocrysts, suggesting a very short duration of the presence of water. This is most naturally explained by the hydrothermal process with decreasing temperatures that took place soon after the extrusion near the ridge or rift. Therefore, it is most likely that the CRM of the komatiites from the Barberton and Belingwe greenstone belts was acquired at almost the same time as the formation of the komatiites, i.e., at ca. 3.5 and 2.7 Ga, respectively.

The problem is that the origin of the ChRM now possessed by the komatiite samples is not known. We can offer three possibilities. The first is that it is overprinted TRM acquired during thermal metamorphic events that occurred after serpentinization. This is the

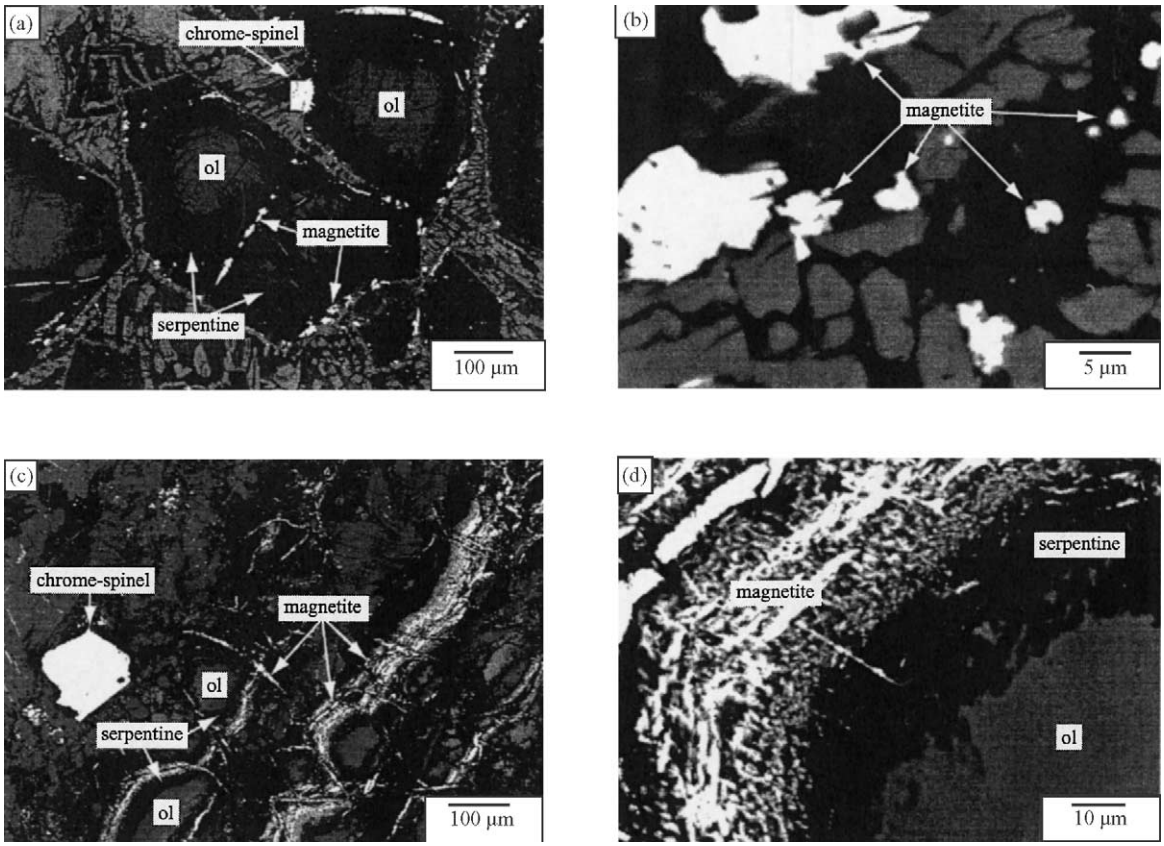


Fig. 16. Electron micrographs of komatiite samples from the Barberton (a, b) and Belingwe (c, d) greenstone belts. ol: olivine.

interpretation by Hale and Dunlop (1984) and Hale (1987b) for the Komati Formation. They considered this thermal overprinting to be associated with the intrusion of the neighboring Theespruit Pluton. The second is that ChRM is thermoviscous remanent magnetization (TVRM) acquired by geologically long-term reheating to a specific temperature during metamorphic events. The third, and the simplest possibility, is that the original grain growth CRM survived as ChRM.

For the type locality area of the Komati Formation in the Barberton greenstone belt, Cloete (1990) and Komiya (1999) reported detailed metamorphic mineralogy of komatiites. According to Cloete (1990), the metamorphic grade is elevated with stratigraphic depth, and amphibolite facies metamorphism is observed at the base of the Komati Formation. However, in the slightly stratigraphically upper (i.e., northern) zone, in which most of the sampling sites for the present study are located (Fig. 3), he reported metamorphic assemblages containing albite and oligoclase in the miscibility gap of peristerite. This means that the metamorphic grade of this zone should be lower than the amphibolite facies. Moreover, he also found the existence of actinolite and hornblende in the miscibility gap in this zone. This is also indicative of metamorphism in this region occurring between the greenschist and epidote-amphibolite facies. Cloete (1990) estimated *P–T* conditions experienced by the komatiites and suggested that the maximum temperature in the sampling area during the metamorphic event was $\leq$500 °C. On the other hand, the greater portion of ChRM possessed by the samples from the Komati Formation resides within a temperature range of 500–590 °C. In particular, more than 50% of the ChRM component in most of the samples lies within the range of 570–590 °C. It therefore is somewhat

difficult to attribute the origin of the ChRM of the Komati samples to overprinted TRM acquired by reheating. When rocks suffer reheating for geologically long periods, some magnetization components in a temperature range higher than the reheating temperature can be replaced with a thermoviscous overprint. According to a diagram of blocking temperature and relaxation time by Pullaiah et al. (1975), the relaxation time at 500 °C (maximum temperature estimate of reheating for the Komati Formation) for SD grains of magnetite with the blocking temperature of 550 °C is as long as 1 Gyr. Dunlop and Buchan (1977) provided an approximate scale of the survival potential of primary NRM for magnetite during thermal metamorphism by referring to this diagram, in which they estimated the minimum blocking temperature of primary NRM for the greenschist facies (extreme metamorphic facies for the Komati samples) to be 500 °C. It hence also appears difficult to regard the entire ranges of the ChRM as TVRM. The overprinted TRM or TVRM components might be equivalent to the middle-temperature components observed in some samples, for their temperature range (typically 300–500 °C) was generally consistent with the temperature condition of metamorphism. Based on the above considerations, consequently, we here propose as the most likely scenario the third possibility, which is that most fractions of ChRM are survivors of the primary grain growth CRMs.

In komatiite from the Belingwe greenstone belt, it is easer to judge the origin of their ChRM, since komatiites occurring in the Reliance Formation are thought to be even fresher than those of the Komati Formation. Metamorphism of the Reliance Formation is usually in the low greenschist or sub-greenschist facies (Bickle et al., 1993), and three locations of sites DS01–02, LS01–03 and NR01–03 are reported to be localities where the rocks are relatively fresh (Martin, 1978). According to Shimizu (personal communication, 2000), the komatiites of these localities have not undergone critical metamorphism, and the temperature of this very low grade metamorphism is estimated to be lower than 200 °C from metamorphic mineral assemblages in basaltic rocks occurring near these localities. In contrast, the ChRM components (i.e., middle-temperature components) of our samples were isolated within the temperature range of 400–500 °C to 570–580 °C, strongly suggesting that they have not suffered any critical overprinting, i.e., that they are grain growth CRMs acquired during serpentinization.

7.2. Thellier–Thellier type paleointensity determination applied to CRM

We will now consider Thellier–Thellier type paleointensity determinations applied to CRM, and implications of the results yielded by the komatiite samples. The behavior of grain growth CRM in SD grains can be predicted by Néel's (1949) theory for the acquisition of TRM. The theoretical comparison between CRM and TRM intensities acquired by the same SD grains was presented by Stacey and Banerjee (1974). Assuming that characteristic relaxation times for CRM and TRM are comparable, the CRM/TRM ratio can be simplified to CRM/TRM $= H_c(T_b)/H_c(T_a)$. Here, T_b and T_a are the blocking temperature of the magnetic grains and the ambient temperature at which their CRM acquisition occurs, respectively. Since $H_c(T)$ for magnetite is virtually proportional to $M_s(T)$, CRM acquired during grain growth due to chemical change occurring at a relatively low temperature ($T_a < T_b$), such as hydrothermal alteration, should always be smaller than TRM. Stacey and Banerjee (1974) quoted a typical numerical value of the CRM/TRM ratio as 0.4.

Several experimental comparisons between CRM and TRM intensities have been published. Pucher (1969) compared demagnetization curves of CRM and TRM using a sample in which pure magnetite grains of 0.2–3 μm in size were grown by the hydrothermal high-pressure method. The TRM intensity produced at 40 μT was six times stronger than the CRM acquired by the same sample. Stokking and Tauxe (1990) synthesized SD hematite in various applied magnetic fields and demonstrated that CRM intensity was linearly related to applied field intensity up to 0.8 mT. They performed simplified Thellier–Thellier type paleointensity experiments and found a reasonably consistent CRM/TRM ratio of 0.15 when using data only from temperature intervals up to 500 °C that showed a linear relationship between CRM removed and pTRM gained. These two experimental estimates are even smaller than the theoretical prediction of Stacey and Banerjee (1974).

In their analysis, Stacey and Banerjee (1974) employed an idealization that magnetic crystals all attained an identical final volume and that the char-

acteristic relaxation times for CRM and TRM are comparable. In reality, however, it is likely that grains grow into various sizes above their blocking volume, creating a blocking temperature spectrum. Hence, the CRM/TRM ratio will not be constant throughout the blocking temperature range. McClelland (1996) calculated the variation of the CRM/TRM ratio with blocking temperatures for magnetite and hematite, using the standard equation for relaxation time by Néel (1949) and the experimental values of spontaneous magnetization as a function of temperature. For magnetite, first, parameters such as blocking volume, blocking temperature, and maximum volume reached were evaluated for fast-grown grains at the ambient temperature of 100 °C over the grain growth timescales of 10^3 s and slow-grown grains over 10^5 years, and for modeling Pucher's (1969) experiments (at 300 °C over 10^5 s). Then, the CRM/TRM ratio at each blocking temperature interval was calculated. The most important point of the results is that the calculated CRM/TRM ratios continuously decreased with increasing temperature. In the case of the fast-grown magnetites, the ratio decreased from 1.0 at 100 °C to 0.2 at 555 °C. The results for the conditions of Pucher's (1969) experiments were also close to this case. With slow-grown grains, the CRM was considerably enhanced compared to the TRM in the low blocking temperature range, with a ratio to TRM of 1.5 at 250 °C. The partial CRM of the high-blocking temperature intervals, however, was much smaller than the partial TRM of the same intervals, with ratios decreasing from 0.8 at 505 °C to 0.3 at 555 °C. McClelland (1996) argues that total intensity of a sample dominated by high blocking temperature grains will be dominated by the contribution from those grains, which will be much stronger for TRM than CRM.

Qualitatively, therefore, Thellier–Thellier type experiments applied to CRM carried by SD magnetites with high blocking temperatures can yield apparently smaller values than the actual field intensities. While the prediction by McClelland (1996) showed variable CRM/TRM ratios with temperature, the data points used for the determinations in many NRM–TRM diagrams of the present study showed reasonable linearity at certain very high temperature intervals. As summarized in Table 7, for almost all of the Barberton samples (20 of 23) except for sites BT09 and TV01, the linear segments were taken from temperature ranges as narrow as 20–30 °C between 560 and 590 °C. Likewise, the typical ranges of the linear relationships for the Belingwe samples (24 of 29) were 30–40 °C between 510 and 570 °C as tabulated in Table 8. With such narrow temperature intervals, the blocking temperature spectra for CRM and TRM could be similar to each other, i.e., the CRM/TRM ratio for an individual sample could be regarded as virtually constant, so that a straight line could be fitted under the criteria. It should be emphasized, however, that the above theoretical models are valid only for the case of CRM possessed by an assemblage of non-interacting SD grains. While the ChRM of the samples is presumed to be carried mainly by SD grains or very small grains akin to SD, the interaction between particles might have affected CRM acquisition, as the magnetite concentrations of the samples are fairly high (see Section 4).

7.3. Constraints on the geomagnetic field intensities at 3.5 and 2.7 Ga

The paleointensity results obtained from the komatiite samples were very similar between the Barberton and Belingwe greenstone belts, yielding very weak formation-mean values calculated as VDMs of $(1.8 \pm 1.3) \times 10^{22}$ A m^2 and $(1.1 \pm 0.9) \times 10^{22}$ A m^2, respectively. These correspond to 24 and 15% of the present dipole moment, and can be taken as being almost the same as the lowest value of the typical dipole fluctuation during the Phanerozoic times (Prévot and Perrin, 1992) or even slightly lower. This might reflect the similarity of the distinctive origin of the magnetic carriers between the two formations. Hale (1987b) reported that komatiites from the Komati Formation yielded a mean equatorial value of 5 μT, which is 17% of the present field intensity. This low value itself seems to be consistent with our results for Barberton, although the temperature ranges of the linear segments used by Hale (1987b) are much wider and lower than those of this study.

In this section, it has been postulated that the most likely origin of most fractions of komatiite ChRM for both greenstone belts is the primary grain growth CRM. As summarized in Tables 7 and 8, the pa-

leointensity results were dispersed to some extent between sites and even within sites. The site-mean VDMs for Barberton ranged from 0.7×10^{22} A m^2 to 4.8×10^{22} A m^2, corresponding to 8–62% of the present field intensity. For the Belingwe samples, the range of the mean VDMs was 0.3×10^{22} A m^2 to 2.5×10^{22} A m^2, corresponding to 4–32% of the present value. The dispersions could be explained by differences between lava flows and even between samples, in composition of melt, degree of serpentinization, ambient temperature during alteration, typical grain size of magnetization carriers, etc., which could affect CRM/TRM ratios, rather than by dipole fluctuation. However, the possibility that some zones in the sampling areas have partially suffered thermal overprinting to a much higher degree than others cannot be denied completely. There seems to be a certain correlation between the paleointensity values and temperature intervals used for linear fittings. The highest mean paleointensity for Barberton was obtained from site BT09, which is the only site where temperature ranges of the linear segments extended down to 400 or 350 °C. As for site SV04, the only specimen (SV4022) for which the lowest temperature of the fitting range was as low as 450 °C yielded an intensity approximately two times higher than other samples from the same site. A similar tendency can be seen in the Belingwe sites, NR02 and NR03. This tendency could raise the question of the relationship between these samples with lower and wider temperature ranges of the linear segments and thermal overprinting. There is, however, neither satisfactory evidence that they have acquired genuine TRM nor sufficient grounds for distinguishing samples with overprinted TRM from those with CRM only by temperature ranges.

In considering the possible origin of the komatiite ChRM, therefore, averaging the results of the paleointensity experiments can provide constraints on the lower limits of the geomagnetic field intensities in both greenstone belts at ca. 3.5 and 2.7 Ga. It is of course impossible to deduce the exact field strength from Thellier–Thellier type experiments applied to samples with naturally occurring CRM. However, actual field strength should be slightly to several times larger than the experimental results, especially for samples dominated by high blocking temperature grains, taking the above described theoretical and experimental estimates of the CRM/TRM ratios into account. It is hence most likely that the magnitude of the geomagnetic field of these ages has fallen into a range comparable to the Phanerozoic dipole fluctuation.

With regard to the Archean geomagnetic field, three paleointensity works were published recently besides Hale's (1987b) report. Approximately 2.6 Ga diabase dikes of the Slave Province, Canada, yielded VDM values of $(6.3 \pm 0.2) \times 10^{22}$ A m^2 and $(9.0 \pm 0.2) \times 10^{22}$ A m^2 similar to the present field strength (Yoshihara and Hamano, 2000), and a very low VDM of $(1.9 \pm 0.6) \times 10^{22}$ A m^2 was obtained from a ca. 2.8 Ga dolerite dike of Greenland (Morimoto et al., 1997). An anorthosite sample from the ca. 2.7 Ga Stillwater Complex, whose age is the closest to that of the komatiites of Belingwe, yielded an intermediate VDM of $(4.1 \pm 0.5) \times 10^{22}$ A m^2 (Selkin et al., 2000). The implication that the actual field strength at ca. 2.7 Ga may have been several times larger than the Belingwe results does not conflict with the moderate nature of the VDM from the Stillwater Complex. We can also point out that the picture of moderate field strength at ca. 3.5 Ga within the bounds of the possibility implied from the Barberton results does not contradict the evolution model of geomagnetic field intensity by Stevenson et al. (1983) that the dynamo generation driven by thermal convection in the Earth's core was still active during the early to middle Archean.

8. Conclusions

We have shown detailed paleomagnetic results with Archean komatiites from the Barberton (ca. 3.5 Ga) and Belingwe (ca. 2.7 Ga) greenstone belts. These include the following observations:

1. Thermal demagnetization shows that komatiite samples from both greenstone belts have stable ChRM components. Paleodirectional data for Barberton agree with the previous report (Hale and Dunlop, 1984), and the ChRM of the Belingwe samples reveals a positive fold test.
2. Rock magnetic investigations suggest that the main carriers of ChRM are Ti-free magnetites presumed

to be akin to single domain grains. Their rock magnetic properties are very stable to laboratory heating up to 600–700 °C. SEM observations support that they are secondary minerals formed during serpentinization.

3. Thellier–Thellier paleointensity experiments yield very low formation-mean VDM estimates of $(1.8 \pm 1.3) \times 10^{22}$ A m^2 for Barberton and $(1.1 \pm 0.9) \times 10^{22}$ A m^2 for Belingwe. The low VDM for the Barberton komatiites is consistent with the previous report by Hale (1987b).

Based on the above observations and consideration of the low metamorphic grade for the sampling areas, we have proposed as the most likely scenario the possibility that most fractions of the komatiite ChRM are survivors of the primary grain growth CRM rather than thermal overprints. Taking the theoretical and experimental estimates of CRM/TRM ratios into account, this suggests that these low mean VDM values obtained can provide constraints on the lower limits of the geomagnetic field intensities in both greenstone belts at ca. 3.5 and 2.7 Ga.

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